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(54) **STATE RETENTION LOAD CONTROL SYSTEM**

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H02J 3/00; **H01H 9/54**
See application file for complete search history.

(56) **References Cited**

U.S. PATENT DOCUMENTS

5,248,919 A 9/1993 Hanna et al.
5,264,761 A 11/1993 Johnson et al.
(Continued)

FOREIGN PATENT DOCUMENTS

CN 2596671 Y 12/2003
CN 101347050 A 1/2009

(Continued)

Primary Examiner — Wei (Victor) Y Chan

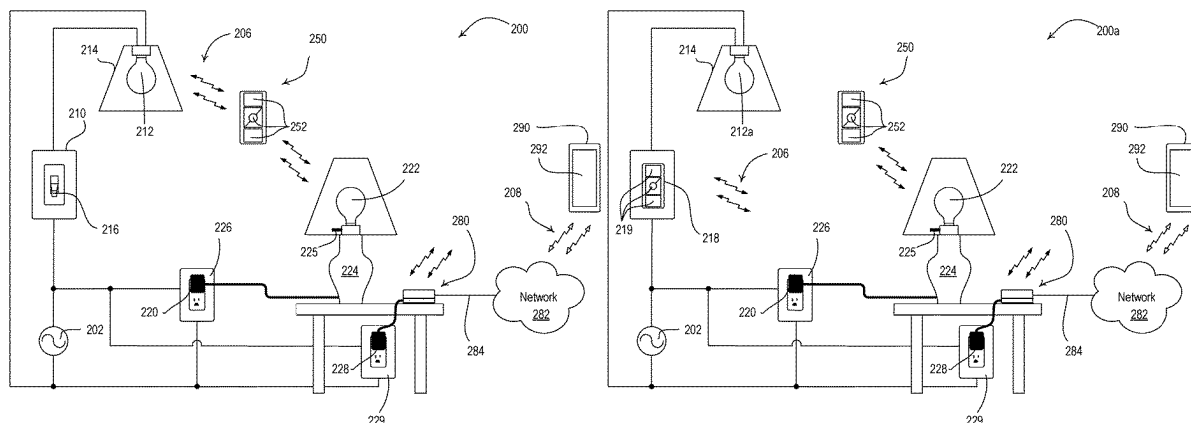
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(57)

ABSTRACT

A load control system may include control devices for controlling power provided to an electrical load. The control devices may include an input device and a load control device. The load control system may include a hub device. The hub device may include a communication circuit and a control circuit. The communication circuit may be configured to receive a digital message from the control device. The control circuit may be configured to determine, based on content of the digital message, whether the control device has experienced a power removal event. The hub device may send, via the communication circuit, a power removal event indication to the control device of whether the control device has experienced the power removal event.

31 Claims, 12 Drawing Sheets



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- continuation of application No. 17/521,015, filed on Nov. 8, 2021, now Pat. No. 11,658,840, which is a continuation of application No. 16/945,462, filed on Jul. 31, 2020, now Pat. No. 11,196,581, which is a continuation of application No. 15/641,933, filed on Jul. 5, 2017, now Pat. No. 10,772,180.
- (60) Provisional application No. 62/358,435, filed on Jul. 5, 2016.
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(56) References Cited**U.S. PATENT DOCUMENTS**

5,905,442 A 5/1999 Mosebrook et al.
 6,803,728 B2 10/2004 Balasubramaniam et al.
 7,190,126 B1 3/2007 Paton
 7,834,856 B2 11/2010 Grinshpoon et al.
 7,936,281 B2 5/2011 Rigatti et al.
 8,009,042 B2 8/2011 Steiner et al.
 8,167,457 B1 5/2012 Forster et al.
 8,199,010 B2 6/2012 Sloan et al.
 8,228,184 B2 7/2012 Blakeley et al.
 8,305,014 B1 11/2012 Li et al.
 8,330,638 B2 12/2012 Altonen et al.
 8,410,706 B2 4/2013 Steiner et al.
 8,451,116 B2 5/2013 Steiner et al.
 8,912,735 B2 12/2014 Chobot et al.
 8,950,461 B2 2/2015 Adams et al.
 9,115,537 B2 8/2015 Blair
 9,208,965 B2 12/2015 Busby et al.
 9,368,025 B2 6/2016 Carmen
 9,374,874 B1 6/2016 Ewing
 D761,277 S 7/2016 Harvell
 9,418,802 B2 8/2016 Romano et al.
 9,520,247 B1 12/2016 Finnegan et al.
 9,583,288 B2 2/2017 Jones et al.
 9,615,434 B2 4/2017 Vangeel et al.
 9,633,557 B2 4/2017 Twaddell et al.
 9,679,696 B2 6/2017 Bhutani et al.
 9,746,138 B1 8/2017 Thomas et al.
 9,799,469 B2 10/2017 Bailey et al.
 9,832,848 B2 11/2017 Hidaka
 9,851,735 B2 12/2017 Dolan et al.
 9,907,138 B2 2/2018 Wang et al.
 9,959,997 B2 5/2018 Bailey et al.
 9,961,750 B2 5/2018 Leinen et al.
 9,967,952 B2 5/2018 Pandharipande et al.
 10,027,127 B2 7/2018 Crafts et al.
 10,225,903 B1 3/2019 Fernandez
 10,271,407 B2 4/2019 Pessina et al.
 2003/0019733 A1 1/2003 Sato
 2003/0019734 A1 1/2003 Sato et al.
 2003/0230982 A1 12/2003 Pagano et al.
 2007/0273509 A1 11/2007 Gananathan
 2008/0067871 A1 3/2008 Black et al.

2008/0111491 A1 5/2008 Spira et al.
 2008/0315798 A1 12/2008 Diederiks et al.
 2009/0167196 A1 7/2009 Wang et al.
 2009/0206983 A1 8/2009 Knode et al.
 2009/0256483 A1 10/2009 Gehman et al.
 2009/0302782 A1 12/2009 Smith et al.
 2010/0084996 A1 4/2010 Van De Sluis et al.
 2010/0127626 A1 5/2010 Altonen et al.
 2010/0244706 A1 9/2010 Steiner et al.
 2010/0244709 A1 9/2010 Steiner et al.
 2011/0314163 A1 12/2011 Borins et al.
 2012/0013257 A1 1/2012 Sibert
 2012/0043889 A1 2/2012 Recker et al.
 2012/0242231 A1 9/2012 Yianni et al.
 2012/0261078 A1 10/2012 Adams et al.
 2012/0286940 A1 11/2012 Carmen et al.
 2012/0292174 A1 11/2012 Mah et al.
 2013/0030589 A1 1/2013 Pessina et al.
 2013/0099011 A1 4/2013 Matsuoka et al.
 2013/0099124 A1 4/2013 Filson et al.
 2013/0214609 A1 8/2013 Carmen et al.
 2013/0338839 A1 12/2013 Rogers et al.
 2014/0001977 A1 1/2014 Zacharchuk et al.
 2014/0117859 A1 5/2014 Swatsky et al.
 2014/0117871 A1* 5/2014 Swatsky H05B 47/11 315/246

2014/0132475 A1 5/2014 Bhutani et al.
 2014/0145646 A1 5/2014 Zhang et al.
 2014/0152186 A1 6/2014 Zhang
 2014/0159588 A1 6/2014 So
 2014/0177469 A1 6/2014 Neyhart
 2014/0180486 A1 6/2014 Newman et al.
 2014/0180487 A1 6/2014 Bull
 2014/0231032 A1 8/2014 Blair et al.
 2014/0265568 A1 9/2014 Crafts et al.
 2014/0265881 A1 9/2014 Karc et al.
 2014/0265918 A1 9/2014 Cummings et al.
 2014/0266669 A1 9/2014 Fadell et al.
 2015/0005900 A1 1/2015 Steele et al.
 2015/0051717 A1 2/2015 Focht et al.
 2015/0077021 A1 3/2015 McCarthy et al.
 2015/0084547 A1 3/2015 Yeh et al.
 2015/0185751 A1 7/2015 Crafts et al.
 2015/0185752 A1 7/2015 Riedl et al.
 2015/0256028 A1 9/2015 Suman
 2015/0294816 A1 10/2015 Evers et al.
 2016/0007431 A1 1/2016 Bosua et al.
 2016/0073479 A1 3/2016 Erchak et al.
 2016/0255698 A1 9/2016 Harbers et al.
 2016/0330825 A1 11/2016 Recker et al.
 2016/0353559 A1 12/2016 Mann et al.
 2016/0360597 A1 12/2016 Greene
 2017/0105176 A1 4/2017 Finnegan et al.
 2017/0223807 A1 8/2017 Recker et al.
 2017/0223808 A1 8/2017 Barna et al.
 2017/0226799 A1 8/2017 Hebeisen et al.
 2017/0279300 A1 9/2017 Catalano et al.
 2018/0014386 A1 1/2018 Bhutani et al.
 2018/0043130 A1 2/2018 Moore-Ede et al.
 2018/0190451 A1 7/2018 Scruggs

FOREIGN PATENT DOCUMENTS

CN 102293060 A 12/2011
 CN 203233569 U 10/2013
 CN 103476186 A 12/2013
 CN 203491998 U 3/2014
 CN 103857146 A 6/2014
 CN 103906323 A 7/2014
 CN 104039035 A 9/2014
 CN 107949127 A 4/2018
 EP 2579691 A1 4/2013
 WO 2014066269 A1 5/2014
 WO 2018078027 A1 5/2018

* cited by examiner

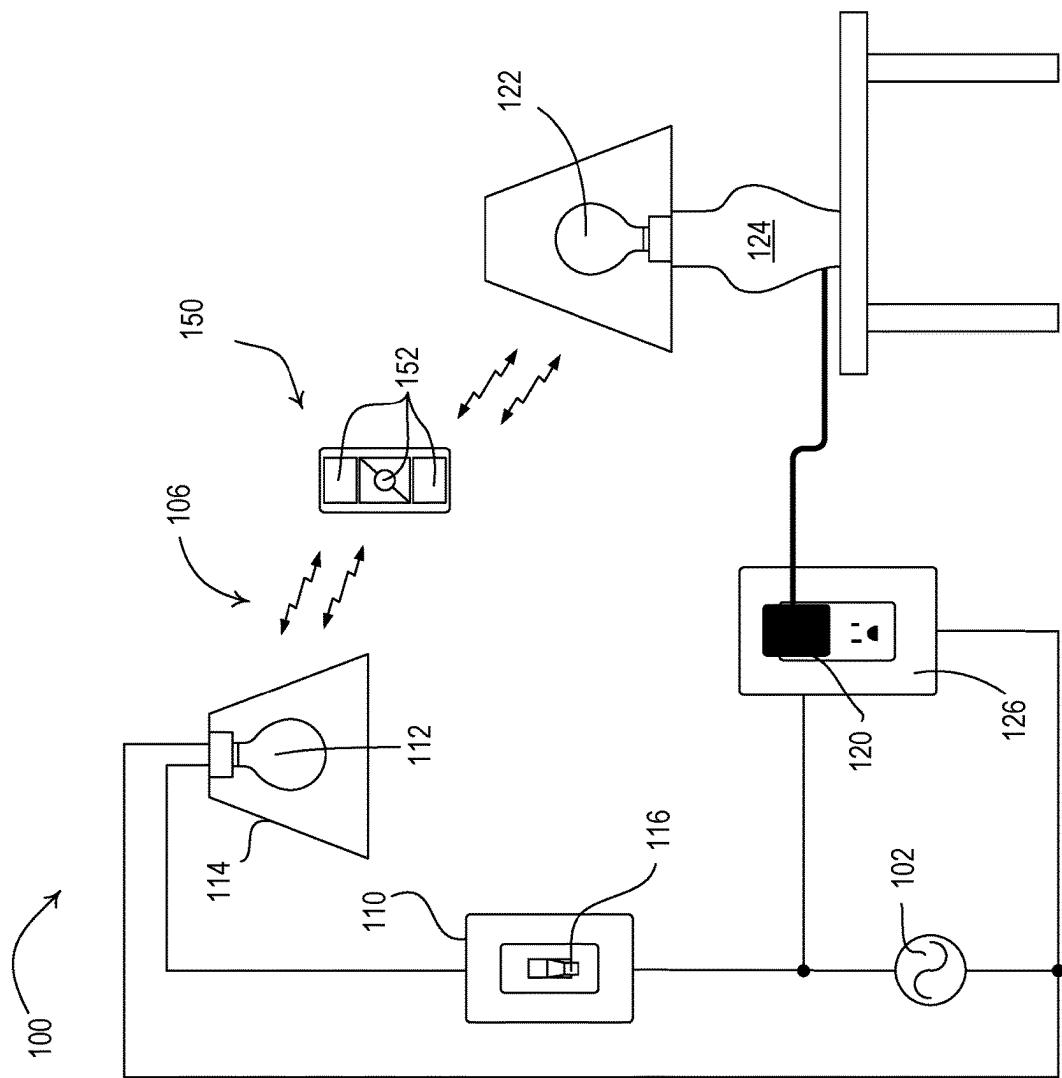


FIG. 1
Prior Art

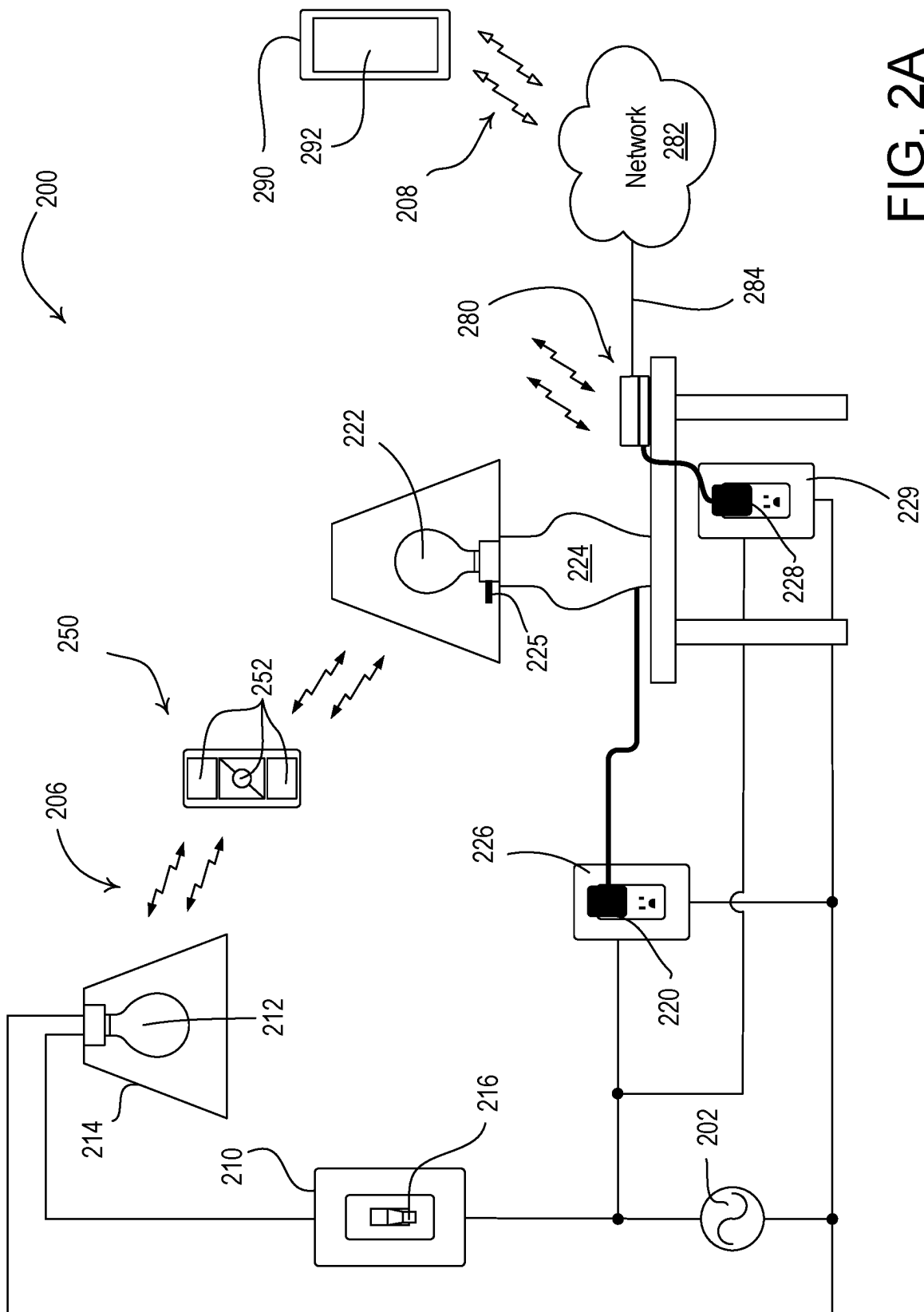


FIG. 2A

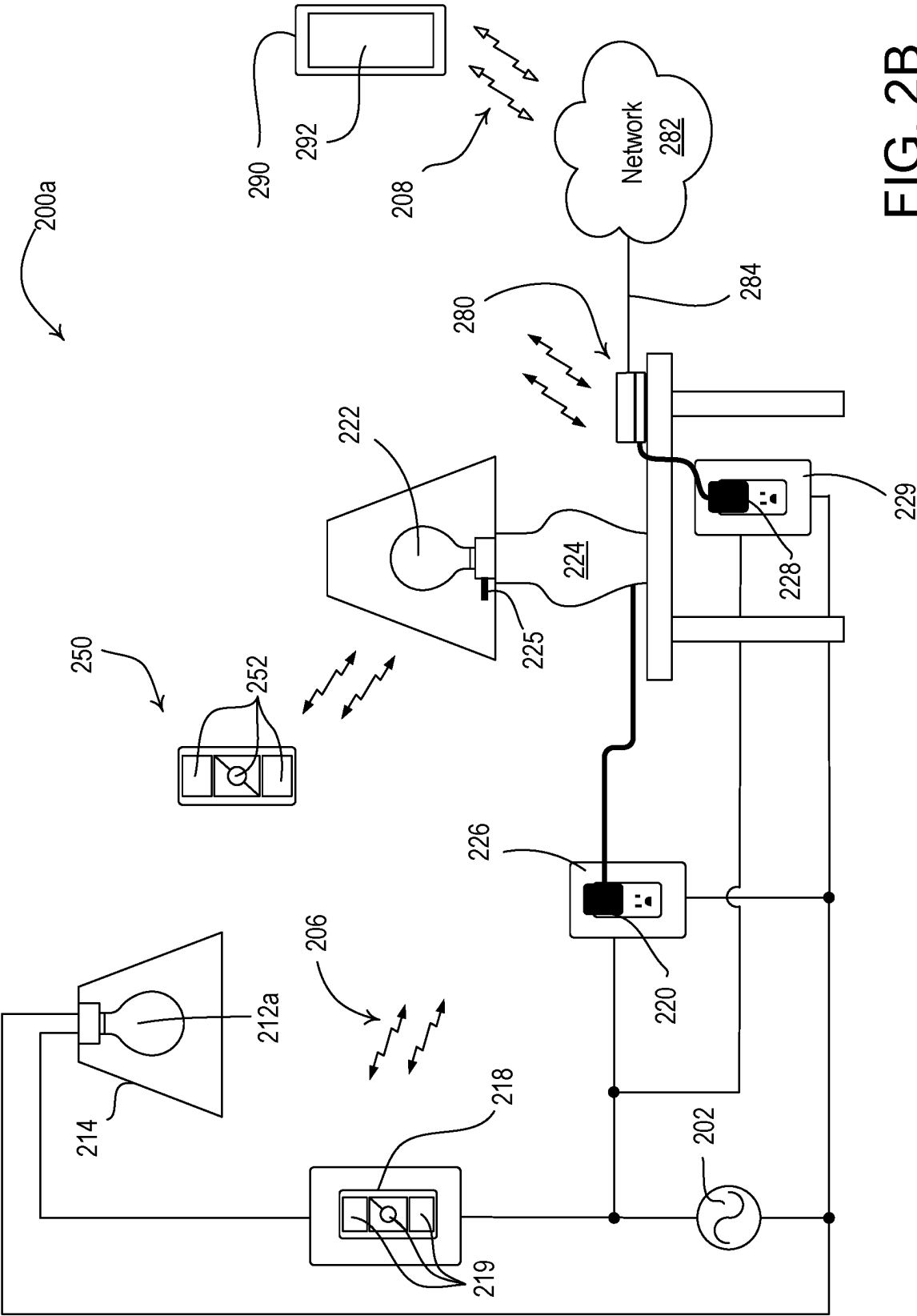


FIG. 2B

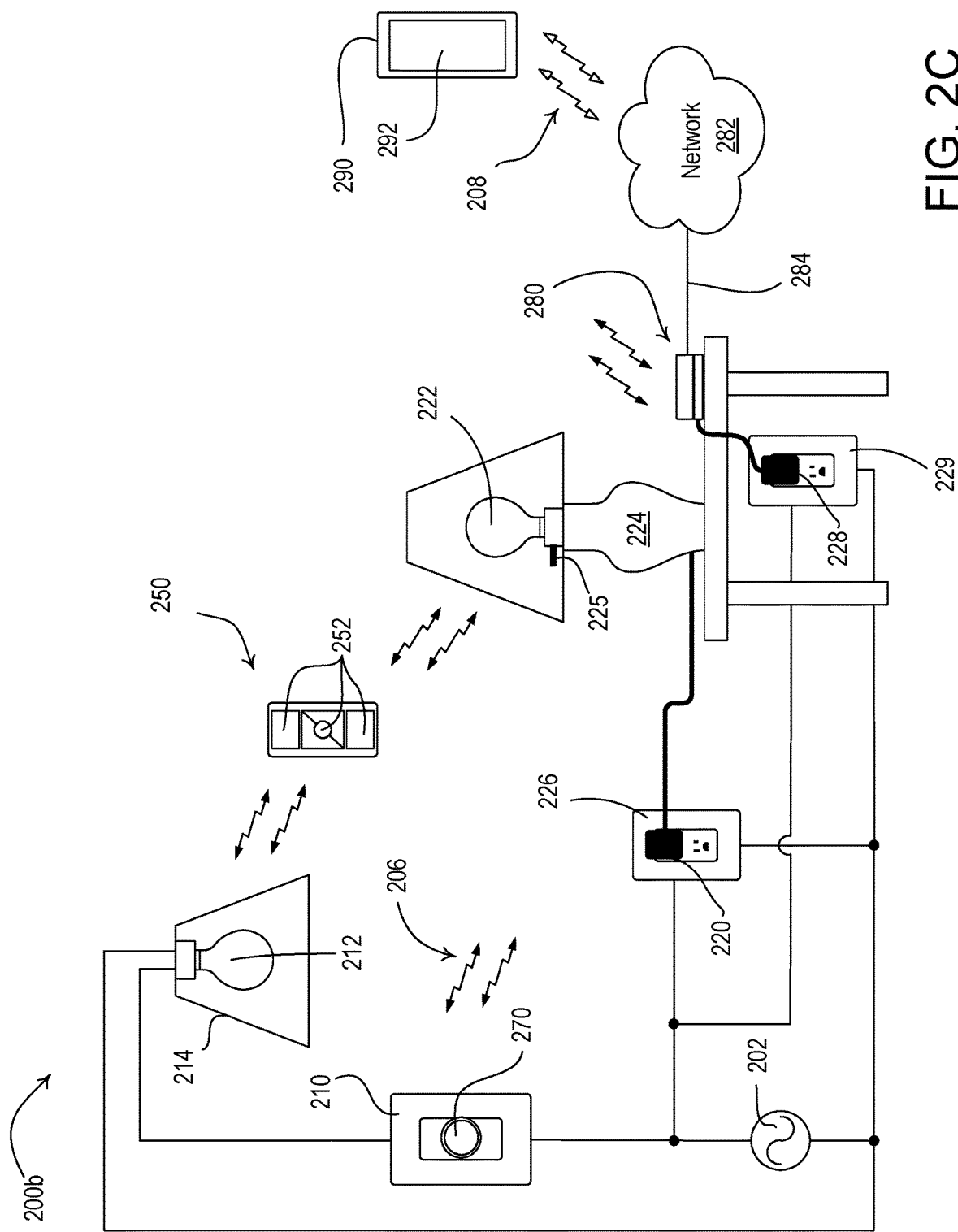
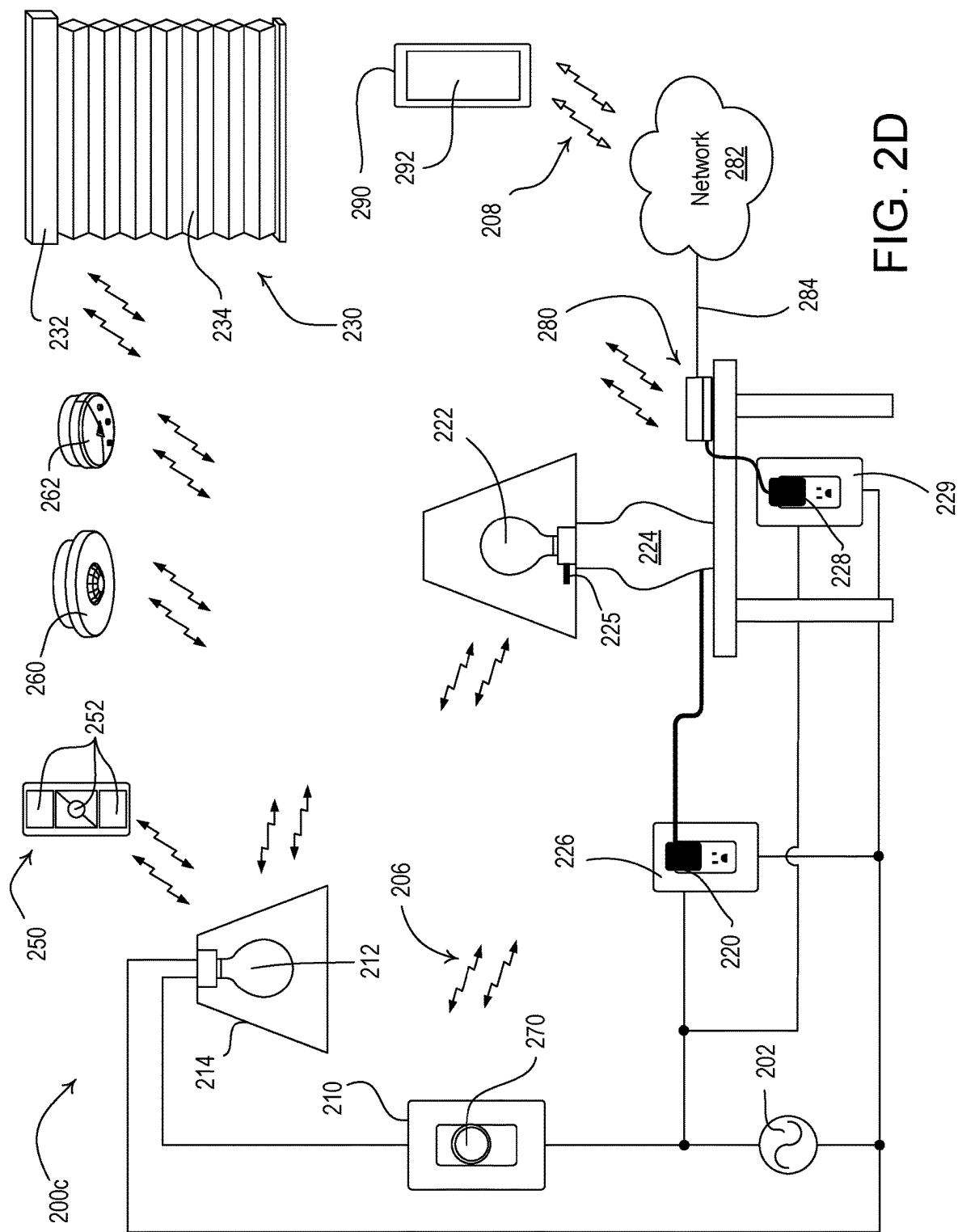


FIG. 2C



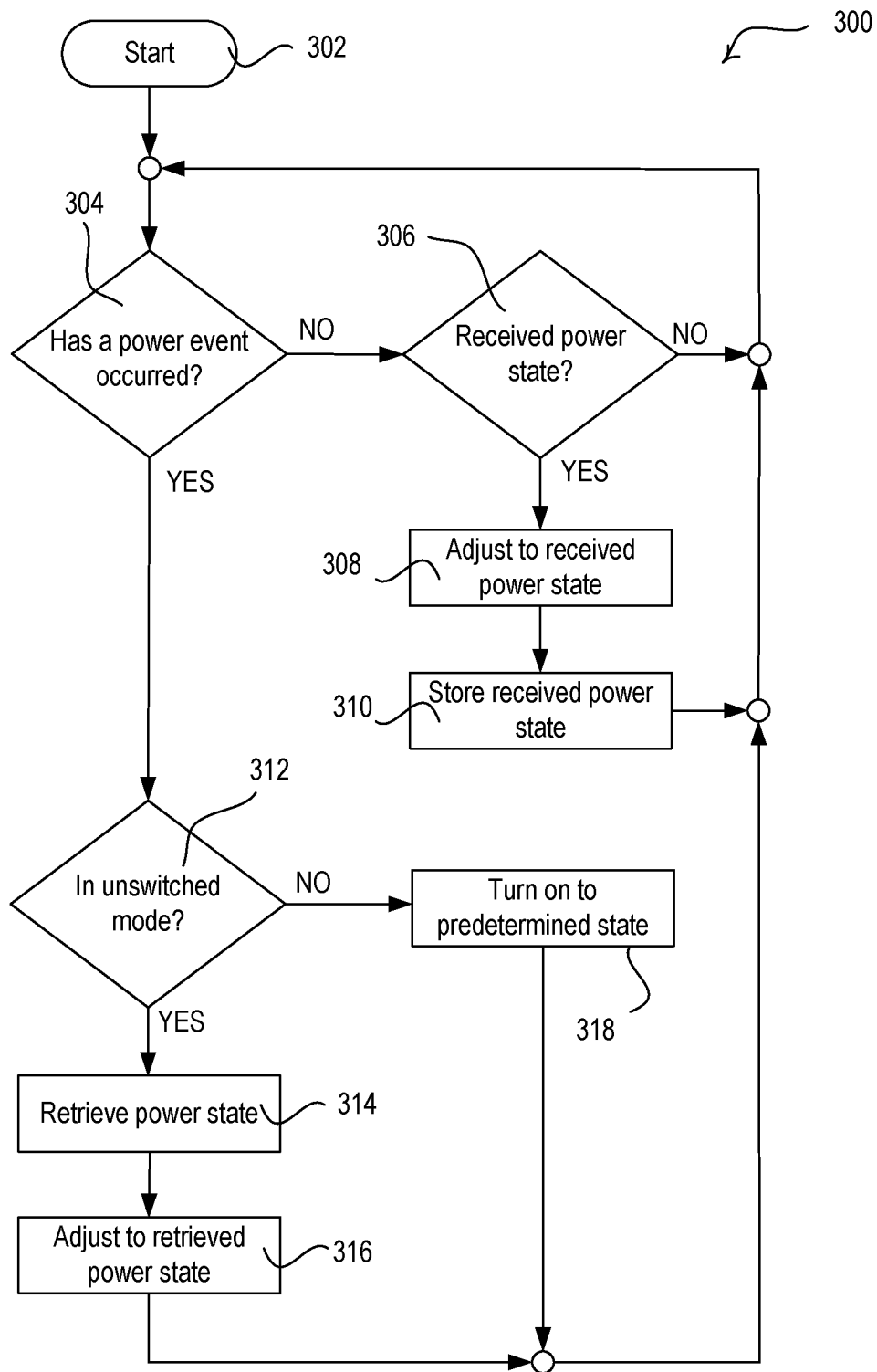


FIG. 3

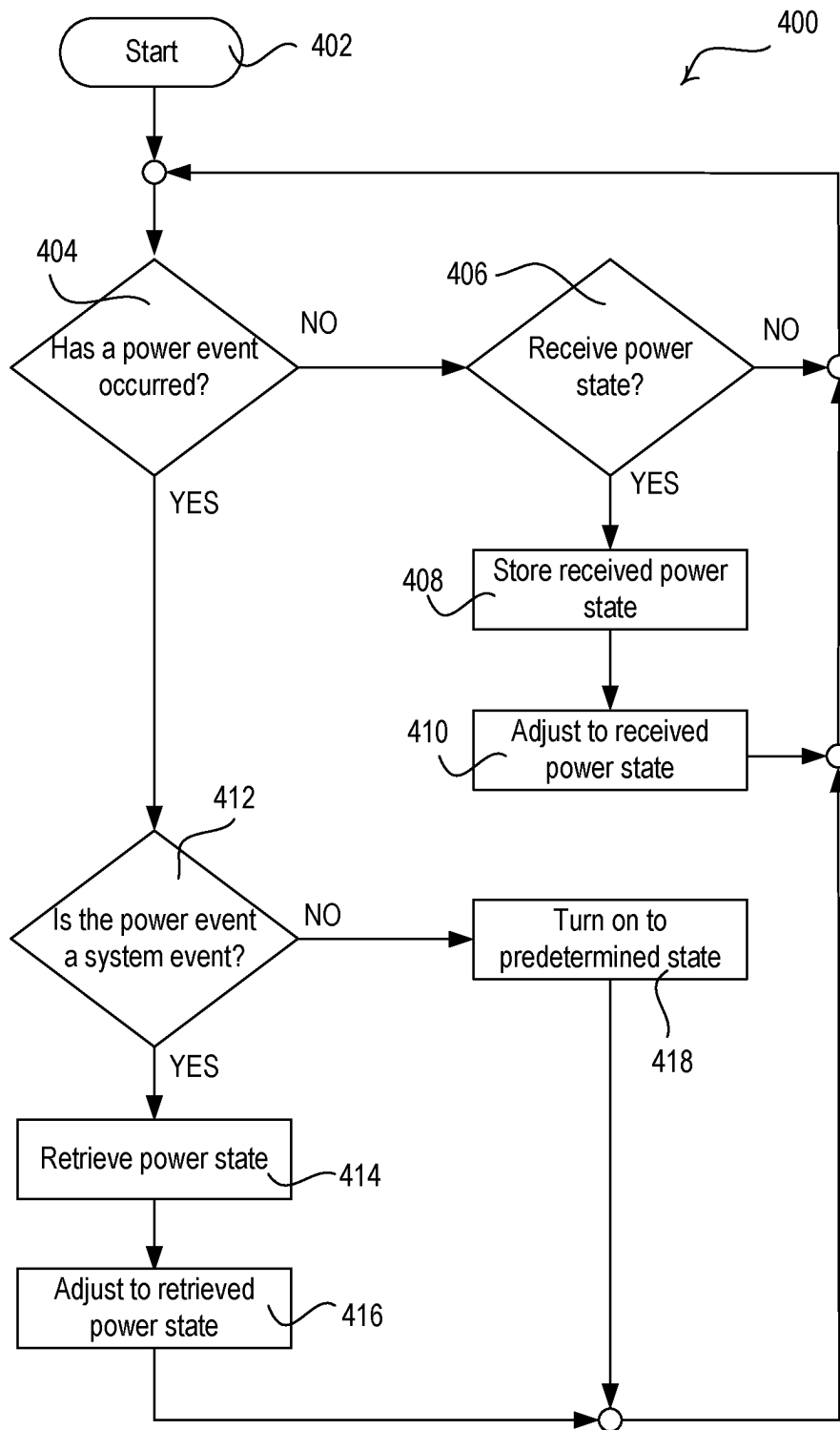


FIG. 4

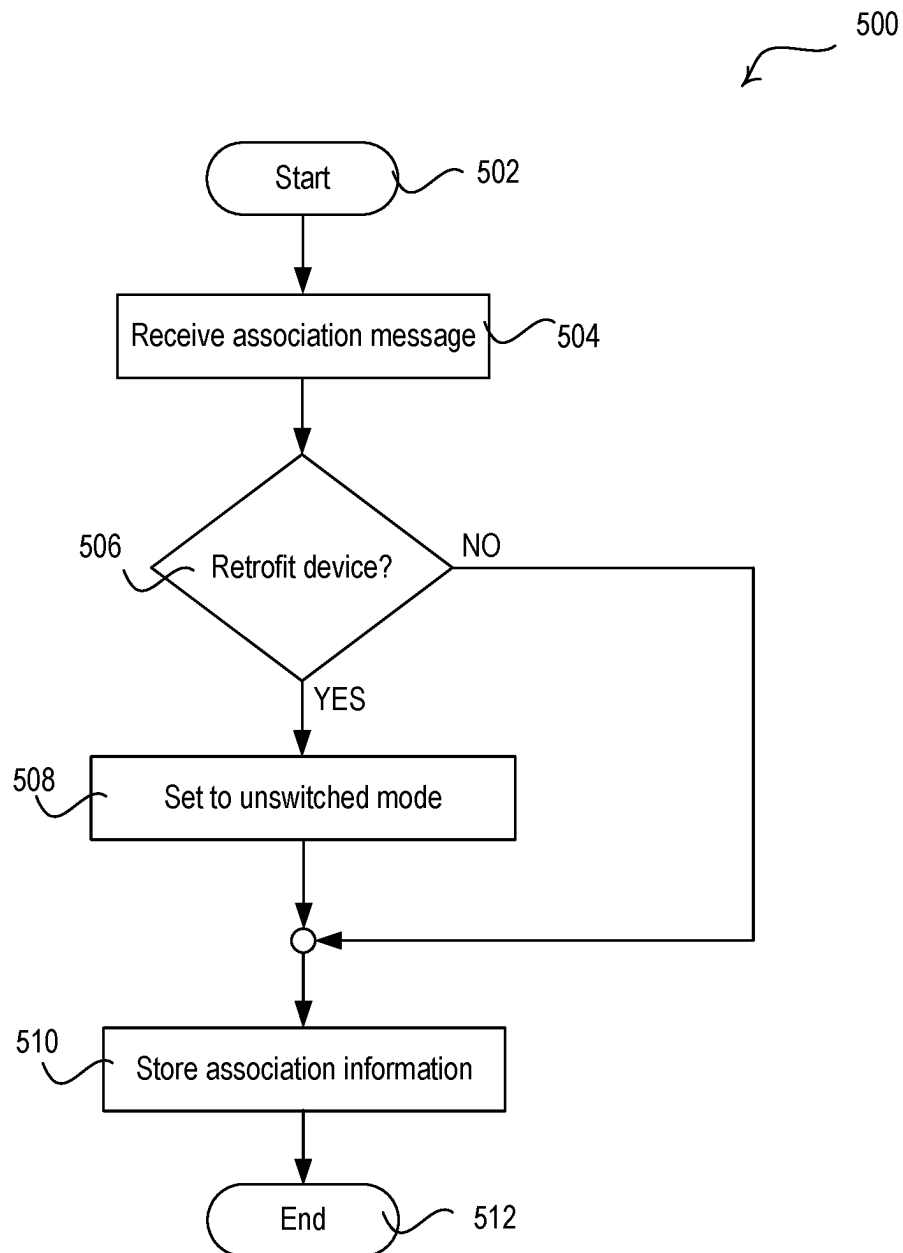


FIG. 5

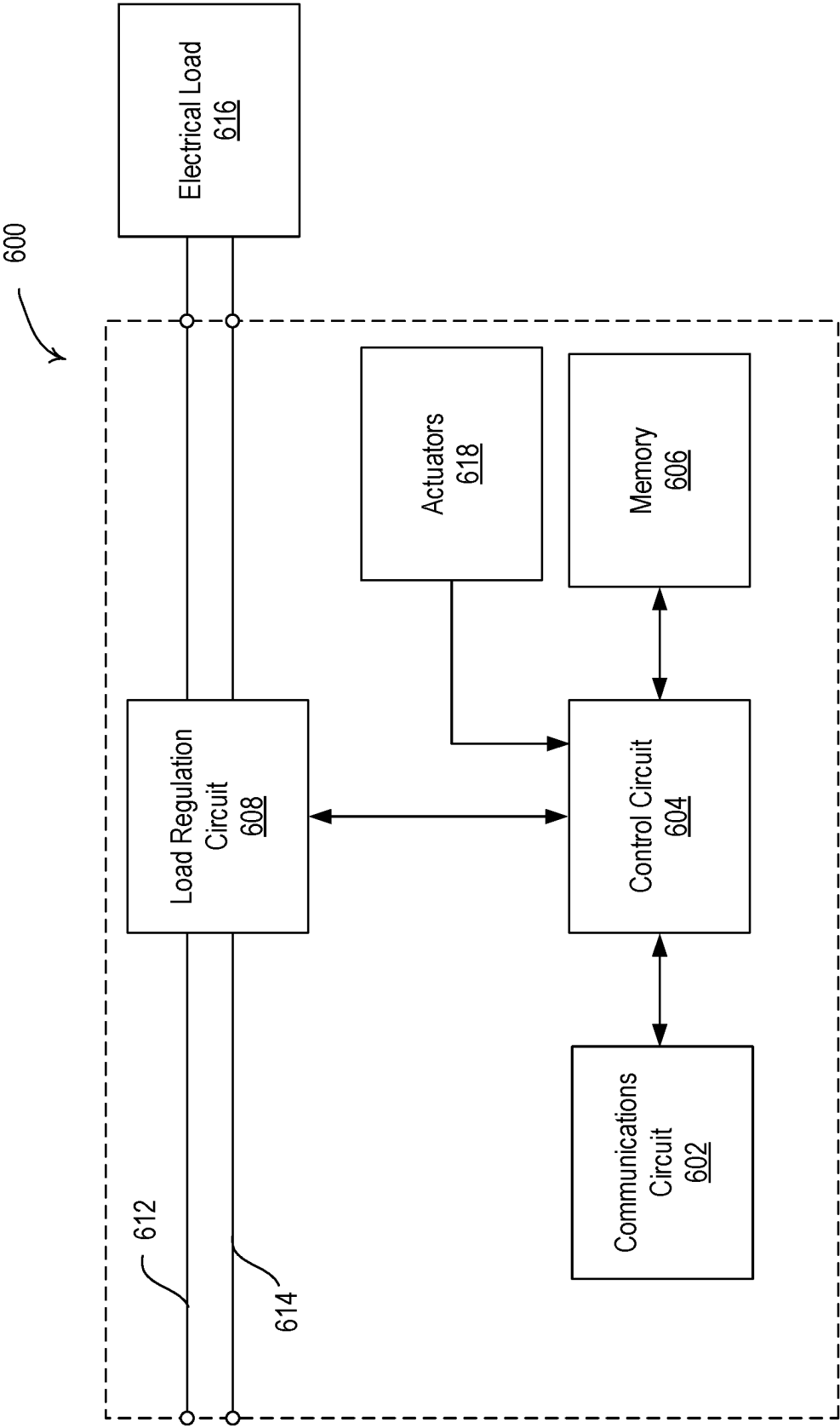


FIG. 6

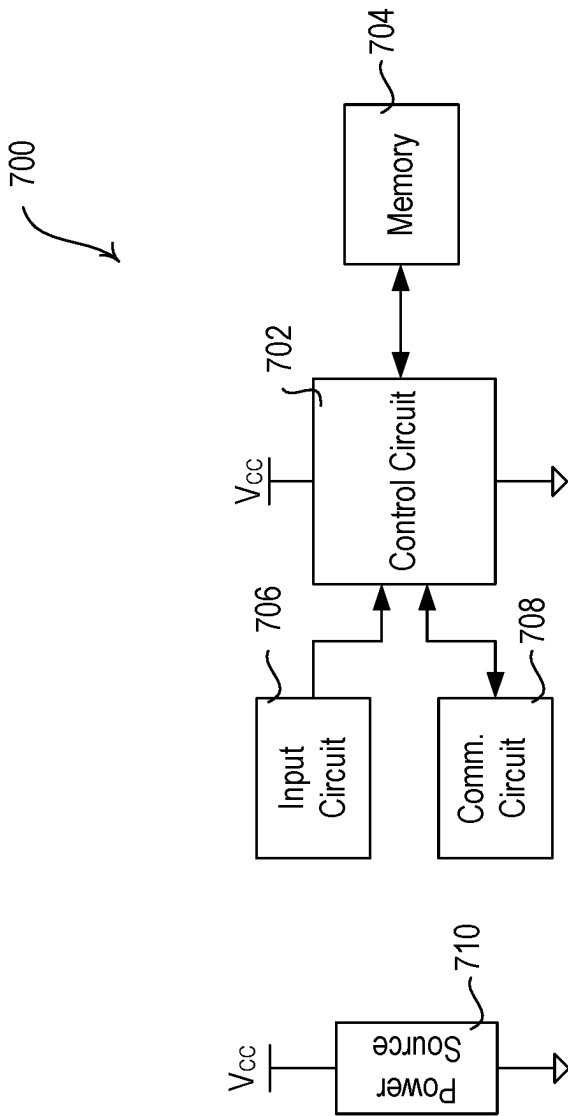


FIG. 7

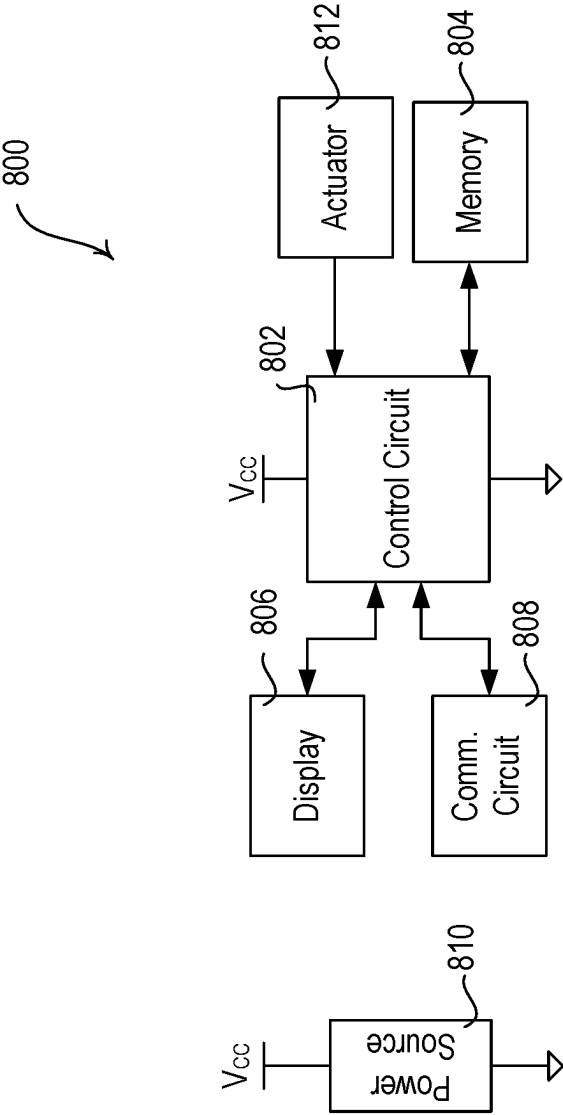


FIG. 8

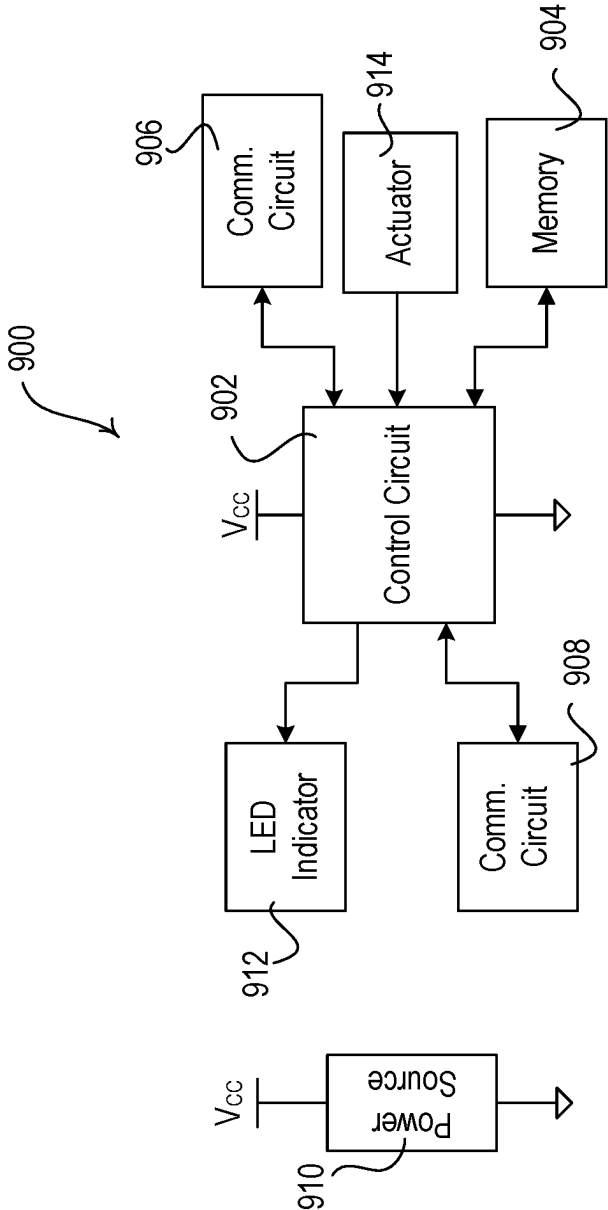


FIG. 9

STATE RETENTION LOAD CONTROL
SYSTEMCROSS-REFERENCE TO RELATED
APPLICATIONS

This application is a continuation of U.S. patent application Ser. No. 18/116,943, filed Mar. 3, 2023, which is a continuation of U.S. patent application Ser. No. 17/521,015, filed Nov. 8, 2021, which issued as U.S. Pat. No. 11,658,840 on May 23, 2023, which is a continuation of U.S. patent application Ser. No. 16/945,462, filed Jul. 31, 2020, which issued as U.S. Pat. No. 11,196,581 on Dec. 7, 2021. U.S. patent application Ser. No. 16/945,462 is a continuation of U.S. patent application Ser. No. 15/641,933, filed Jul. 5, 2017, which issued as U.S. Pat. No. 10,772,180 on Sep. 8, 2020. U.S. patent application Ser. No. 15/641,933, claims the benefit of U.S. Provisional Patent Application No. 62/358,435, filed Jul. 5, 2016. Each of U.S. patent application Ser. No. 18/116,943, U.S. patent application Ser. No. 17/521,015, U.S. patent application Ser. No. 16/945,462, U.S. patent application Ser. No. 15/641,933, U.S. Provisional Patent Application No. 62/358,435, U.S. Pat. Nos. 11,658,840, 11,196,581, and 10,772,180, is incorporated by reference herein in its entirety.

BACKGROUND

A user environment, such as a residence or an office building, for example, may be configured using various types of load control systems. FIG. 1 depicts a prior art user environment **100**. The user environment **100** includes a wall-mounted load control device **110** coupled in series electrical connection between an AC power source **102** and a light bulb **112** installed in a ceiling mounted downlight fixture **114**. The user environment **100** includes a table lamp **124** plugged into an electrical receptacle **126** that is powered by the AC power source **102**. The table lamp **124** is plugged into an electrical receptacle **126** via electrical plug **120**. Light bulb **122** is installed in table lamp **124**. The wall-mounted load control device **110** has an electronic switch (e.g., light switch **116**). In response to actuation of the light switch **116**, the wall-mounted load control device **110** is configured to turn the light bulb **112** on and off.

The user environment **100** includes a battery-powered handheld remote control device **150** having buttons **152**. The battery-powered remote control device **150** transmits RF signals **106** in response to actuations of one or more of the buttons **152**. The light bulb **112** and light bulb **122** receive digital messages via radio-frequency (RF) signals **106**. The RF signals **106** are transmitted by the battery-powered remote control device **150**. In response to the received RF signals, the light bulb **112** and/or light bulb **122** turn on and off.

The user environment may include a bridge and a smartphone or tablet for controlling the light bulbs **112**, **122**. The bridge transmits RF signals to the light bulbs **112**, **122**. The bridge receives Wi-Fi signals from the smartphone or tablet for controlling the light bulbs **112**, **122** and formats the information in the Wi-Fi signals for being received by the light bulbs **112**, **122** on a different protocol.

In user environment **100**, a problem arises if power is removed from the light bulb **112** and/or the light bulb **122**. The problem arises whether or not a bridge is present in the user environment **100**. For example, if power is removed from the light bulbs and returns to the light bulbs, the light bulbs turn on to a default intensity level (e.g., 100% intensity

level). As an example, the light bulb **112** may be set to electronic off and the light bulb **122** may be set to an intensity level of eighty percent. Power may intentionally be removed from the light bulb **112** via an actuation of the light switch **116** to the off position. As another example, power may unintentionally be removed from the light bulb **112** and/or the light bulb **122** due a blackout or a brownout. Upon power being returned to the light bulb **112** and the light bulb **122**, the light bulb **112** and light bulb **122** may present light at the default intensity (e.g., an intensity level of 100%). During unintentional power removal, it may be undesirable to a user for the light bulb **112** and light bulb **122** to present light at the default intensity. Rather, during an unintentional power removal, the user may desire that the intensity of the light bulbs be returned to their respective pre-power removal states (e.g., electronic off and eighty percent, respectively). For example, during an unintentional power removal, a user may not desire the light bulb **112** and/or the light bulb **122** to operate at the default state of full intensity when the user is asleep or when the user is away from the user environment **100** for an extended period of time. Accordingly, there is a need for the control system to distinguish between an intentional power removal event (based on a user action) and an unintentional power removal event (such as a blackout), so that the light bulbs may behave in an expected manner.

SUMMARY

A load control system may include control devices for controlling an amount of power provided to an electrical load. The control devices may include an input device and/or a load control device. The load control devices may be capable of directly controlling an electrical load. The input devices may be capable of indirectly controlling the electrical load via the load control device. For example, the load control device may be capable of controlling the amount of power provided to the electrical load based on digital messages received from the input device and/or another device. The digital messages may include load control instructions or another indication that causes the load control device to determine load control instructions for controlling an electrical load.

The control devices may be powered from a power source (e.g., alternating-current (AC) or direct-current (DC) power source). If an inadequate amount of power is provided to the control devices, the control devices may stop functioning. An inadequate amount of power may occur as a result of a power outage (e.g., a blackout, brownout, or other power loss). For example, during a brownout, a control device powered by an AC circuit may be inoperable. Although control devices may receive an inadequate amount of electric power as a result of power outages, control devices may also, or, alternatively, receive an inadequate amount of power in other ways. For example, control devices may receive an inadequate amount of power as a result of a user action, such as the user turning a light switch off that powers control devices or unplugging a power cord that powers the control devices. Power removal events as a result of power outages may be referred to as system power removal events. Power removal events as a result of user actions may be referred to as local power removal events.

A control device may be configured to be controlled to and retain one or more states (e.g., power states). The control device may include memory. The memory may be volatile and/or non-volatile memory. The memory may be used for saving a power state of the control device. A power state of the control devices may include an off power state of the

control device, an on power state of the control device, an intensity state (e.g., an intensity level) of the control device, a color (e.g., a color temperature) state of the control device, etc. For example, the control device may store in memory that the power state of a control device is an intensity state of eighty-percent. The power state may be recalled at a later time. For example, a prior power state (e.g., a power state of a control device prior to a power removal event) may be recalled by the control device after the power removal event ends. For example, after a power removal event, a control device may recall that the prior power state of the control device was a particular intensity level, such as eighty-five percent. The control device may recall (e.g., from memory within the control device) the prior power state of the control device, for example, when the power removal event ends. A lighting device, for example, may be set to forty percent prior to a power removal event. Upon the power removal event occurring and the power removal event ending, the control device may recall (e.g., from the memory within the control device) that the prior power state of the lighting device was forty percent. Thus, upon the power removal event ending, the lighting device may function at an intensity of forty percent.

A hub device may communicate with one or more control devices. For example, a hub device may communicate with control devices via a wireless and/or wired signal. The hub device may provide command instructions (e.g., power on, power off, set to an intensity level, set to a color temperature, etc.) to the control devices. The hub device may be electrically coupled with a power source (e.g. an AC power source). For example, the hub device may be electrically coupled with a power source, and/or the hub device may be configured to determine the status (e.g., on, off, etc.) of electrical power being distributed from the power source in which the hub device is electrically coupled. The hub device may be electrically coupled with a power source that is electrically coupled with control devices. The hub device may determine the status (e.g., on, off, etc.) of electrical power that the power source is providing to the control devices, and/or the hub device may determine the status of electrical power that the power source is providing to the hub device. For example, the hub device may determine if, and/or when, electrical power is no longer being provided by an AC power source to a control device. The hub device may also, or, alternatively, determine if, and when, electrical power is no longer being provided to the hub device.

The hub device may be able to determine whether a power removal event (e.g., a blackout, brownout, loss of power, etc.) is a local power removal event. A local power removal event may occur as a result of a user turning an electrical switch to the off position. For example, a user may turn an electrical switch to an off position in an attempt to turn off the power and/or functionality of a corresponding control device. A local power removal event may include a power removal event occurring to a single control device, and/or a power removal event occurring to one or more control devices located within a predefined location (e.g., one or more control devices located within a room). The hub device may be able to determine whether a power removal event is a system power removal event (e.g., a blackout, brownout, etc.). A system power removal event may include a power removal event occurring to one or more control devices within one or more locations. A system power removal event may include a power removal event occurring to the hub device and one or more control devices within one or more locations. A system power removal event, for example, may include a power removal event occurring to five control

devices located within three rooms. A system power removal event may occur as a result of a blackout, brownout, other losses of power, etc.

The hub device may include memory. The memory of the hub device may be volatile and/or non-volatile memory. The memory may be used for saving a power state of one or more of the control devices with which the hub device communicates. A power state of the control devices may include an off power state of the control devices, an on power state of the control devices, an intensity level of the control device, a color temperature of the control device, etc. For example, the hub device may store in memory that the power state of one of the control devices is functioning at an eighty-percent intensity level, and another one of the control devices may be functioning at a forty percent intensity level. The power states of the control devices may be recallable (e.g., at a later time) by the hub device. For example, in the event of a power removal event (e.g., in the event of a blackout, brownout, other form of power loss, etc.), the hub device may recall the prior power states of one or more of the control devices.

Upon a power removal event, the hub device may recall that the power state of one of the control devices was at an intensity level of eighty-five percent, and the power state of another one of the control devices was at an intensity level of forty percent. Upon the power removal event ending (e.g., the blackout, brownout, other power loss, etc., ending), the hub device may communicate (e.g., communicate to the control devices) the prior power state of the control devices. For example, the hub device may communicate to one of the control devices that the prior power state of the control device was an intensity level of eighty-five percent, and the hub device may communicate to another one of the control devices that the prior power state of that control device was an intensity level of forty percent. After the end of the power removal event, the control devices may function according to the state (e.g., the prior power removal event state) provided to the respective control devices by the hub device. Thus, after power is restored to the control devices, the control devices may turn on to an intensity level of eighty-five percent and forty percent, respectively.

The hub device may communicate (e.g., send a digital message) with control devices according to a predefined dataset. The predefined dataset may be based on a location of the control devices, a predefined signal threshold, a priority of the control devices, etc. For example, upon determining that a power removal event has occurred, the hub device may send signals to control devices located within a location (e.g., located within a bedroom, within a conference room, within an entire building, etc.). Upon determining that a power removal event has occurred, the hub device may also, or, alternatively, broadcast a signal to each control device within a predefined signal threshold. The signal threshold may correspond to the transmitting signal and/or the signal threshold may correspond to a receiving signal.

The hub device may transmit an indication to the control devices of whether a power removal event (e.g., a brownout, blackout, etc.) has occurred. The indication may include whether the power removal event was a local power removal event or whether the power removal event was a system power removal event. The indication of the power removal event may include the time of the power removal event and/or the duration of the power removal event. The hub device may query one or more of the control devices to determine whether the control devices have experienced and/or are experiencing a power removal event. For

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example, the hub device may ask one or more of the control devices whether they are receiving power from an AC power source. The hub device may determine whether the power removal event is a local power removal event or a system power removal event, based on, for example, whether one or more of the control devices have experienced and/or are experiencing a power removal event.

Upon the power removal event ending, the hub device may send information (e.g., power removal event information, power state information, etc.) to the control devices. The hub device may send the information to the control devices based on whether the power removal event was a system power removal event or a local power removal event. For example, the hub device may send power state information to a control device in the event of a system power removal event and/or the hub device may not send power state information to a control device in the event of a local power removal event. The hub device may also, or alternatively, send power state information to a control device, independent of the type of power removal event that has occurred.

A load control device (e.g., wall-mounted load control device) may be coupled to an external device. The external device may be a retrofit device. For example, a wall-mounted load control device may be coupled to a dimmer device, a remote control device, etc. The external device may be configured to retain the power state of control devices (e.g., lighting devices). For example, the external device may be configured to retain the power state of the control devices to an on power state. The external device may be configured to retain an intensity level of a light bulb to a non-zero value (e.g., an intensity level of 1%-100%). The external device may retain the power state of the light bulb to an on power state, for example, by physically maintaining electrical continuity of the circuit (e.g., AC circuit) between an AC power source and the light bulb.

A control device coupled to the external device may not experience a local power removal event. For example, because the external device retains the power state of the control device to an on power state, the control device may not be turned off by a user. Even when the external device is actuated to its lowest state, or electronic off, the light bulb may continue to draw some power (e.g., about the amount of power to provide a 1% intensity level) from the control device. A control device in a control system coupled the external device may experience a system power removal event. For example, a control device (e.g., a light bulb coupled to the external device) may experience a blackout. Because the external device prevents a local power removal event, a power removal event experienced by the control device coupled to the external device may be determined to be a system power removal event.

Another control device may receive electrical power from an electrical receptacle which is not coupled to the external device. Because control device is not coupled to the external device, the power removal event to the control device may be determined to be a local power removal event and/or the power removal event to the control device may be determined to be a system power removal event. To determine whether a power removal event is a local power removal event or a system power removal event, the power states of one or more control devices may be determined. For example, if a first control device is experiencing a power removal event and a second control device is experiencing a power removal event, it may be determined that the power removal event is a system power removal event. If the first control device is experiencing a power removal event and

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control devices positioned within one or more rooms are experiencing a power removal event, it may be determined that the power removal event is a system power removal event. If the first control device is experiencing a power removal event and the second control device is not experiencing a power removal event, and vice-versa, it may be determined that the power removal event is a local power removal event.

Upon the control devices and/or other devices (e.g., hub device) determining that the power removal event is a system power removal event, the control devices and/or other devices may adjust the power state of the control devices upon the power removal event ending. For example, upon the power removal event ending, the control devices and/or other devices may adjust the power state of the control devices to the prior power states of the control devices. For example, upon the control devices and/or other devices determining that the power removal event is a system power removal event, the control devices and/or other devices may obtain the prior power states of the control devices (e.g., via memory). The control devices and/or other devices may send and/or set the power states of the control devices to the obtained prior power states of the control devices.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 is an example prior art environment including load control devices.

FIGS. 2A-2D are perspective views of example environments for adjusting and/or retaining state information of control devices.

FIGS. 3-5 are simplified flowcharts of example methods for adjusting and/or retaining state information of control devices.

FIG. 6 is a block diagram of an example load control device.

FIG. 7 is a block diagram of an example input device.

FIG. 8 is a block diagram of an example network device.

FIG. 9 is a block diagram of an example hub device.

DETAILED DESCRIPTION

FIG. 2A is a diagram of an example load control system **200** (e.g., a lighting control system) in which one or more lighting devices, such as wirelessly-controllable light bulbs **212**, **222** (e.g., lighting loads) may be deployed. The load control system **200** may comprise a wall-mounted load control device **210** (e.g., a mechanical switch, such as, a toggle switch, a paddle switch, a pushbutton switch, a "light switch," or other suitable switch), which may be coupled in series electrical connection between an alternating-current (AC) power source **202** and the first light bulb **212**. The light bulb **212** may be installed in a ceiling-mounted downlight fixture **214**. In addition, the light bulb **212** may be installed in a wall-mounted lighting fixture or other lighting fixture mounted to another surface. The wall-mounted load control device **210** may include an actuator (e.g., a toggle actuator **216**) for controlling the power delivered to the light bulb **212**. In response to actuations of the toggle actuator **216**, the wall-mounted load control device **210** may be configured to turn the light bulb **212** on and off. Because the wall-mounted load control device **210** may be configured to turn the light bulb **212** on and off (e.g., the light bulb **212** is downstream of the wall-mounted load control device **210**), the light bulb **212** may be considered to be a switched electrical device.

The wall-mounted load control device **210** may be adapted to be wall-mounted in a standard electrical wallbox.

The second light bulb **222** may be installed in a lamp (e.g., a table lamp **224**). The table lamp **224** may be plugged into (e.g., via electrical plug **220**) an electrical receptacle **226** that is powered by the AC power source **202**. The light bulb **222** may be installed in the table lamp **224** or other lamp that may be plugged into the electrical receptacle **226**. The table lamp **224** may comprise an internal switch (not shown) coupled in series between the AC power source **202** and the light bulb **222**. The table lamp **224** may also include an actuator **225** for controlling the internal switch to control the power delivered to the light bulb **222**. In response to actuations of the actuator **225** of the table lamp **224**, the internal switch of the table lamp **224** may be configured to turn the light bulb **222** on and off. Because the internal switch of the table lamp **224** may be configured to turn the light bulb **222** on and off (e.g., the light bulb **222** is downstream of the internal switch of the table lamp **224**), the light bulb **222** may be considered to be a switched electrical device.

The load control system **200** may include one or more input devices, e.g., radio-frequency (RF) transmitters, such as a battery-powered handheld remote control device **250**. The light bulb **212** and/or the light bulb **222** may be configured to receive digital messages via wireless signals, e.g., radio-frequency (RF) signals **206** (e.g., ZIGBEE®; NFC; BLUETOOTH®; WI-FI®; or a proprietary communication channel, such as CLEAR CONNECT™, etc.). The wireless signals may be transmitted by the battery-powered remote control device **250**. In response to the received digital messages, the respective light bulbs **212**, **222** may be turned on and off, and/or the intensities of the respective light bulbs **212**, **222** may be increased or decreased. The battery-powered remote control device **250** may include one or more actuators **252** (e.g., one or more of an on button, an off button, a raise button, a lower button, or a preset button). The battery-powered remote control device **250** may transmit the RF signals **206** in response to actuations of one or more of the actuators **252**. The battery-powered remote control device **250** may be handheld. The battery-powered remote control device **250** may be mounted vertically to a wall, or supported on a pedestal to be mounted on a tabletop. Examples of battery-powered remote control devices are described in greater detail in commonly-assigned U.S. Pat. No. 8,330,638, issued Dec. 11, 2012, entitled WIRELESS BATTERY-POWERED REMOTE CONTROL HAVING MULTIPLE MOUNTING MEANS, and U.S. Patent Application Publication No. 2012/0286940, published Nov. 15, 2012, entitled CONTROL DEVICE HAVING A NIGHT-LIGHT, the entire disclosures of which are hereby incorporated by reference.

Digital messages transmitted by the input devices (e.g., the battery-powered remote control device **250**) may include a command and/or identifying information, such as a serial number (e.g., a unique identifier) associated with the transmitting input device. Each of the input devices may be associated with (e.g., assigned to) the light bulb **212** and/or the light bulb **222** during a configuration procedure of the load control system **200**, such that the light bulb **212** and/or the light bulb **222** may be responsive to digital messages transmitted by the input devices via the RF signals **206**. Examples of associating wireless control devices during a configuration procedure are described in greater detail in commonly-assigned U.S. Patent Application Publication No. 2008/0111491, published May 15, 2008, entitled RADIO-FREQUENCY LIGHTING CONTROL SYSTEM, and U.S. Pat. No. 9,368,025, issued Jun. 14, 2016, entitled

TWO-PART LOAD CONTROL SYSTEM MOUNTABLE TO A SINGLE ELECTRICAL WALLBOX, the entire disclosures of which are hereby incorporated by reference.

The load control system **200** may include a hub device **280** configured to enable communication with a network **282**, e.g., a wireless or wired local area network (LAN). The hub device **280** may be connected to a router via a wired digital communication link **284** (e.g., an Ethernet communication link). The router may allow for communication with the network **282**, e.g., for access to the Internet. The hub device **280** may be wirelessly connected to the network **282**, e.g., using wireless technology, such as Wi-Fi technology, cellular technology, etc. The hub device **280** may be configured to transmit communication signals (e.g., RF signals **206**) to the light bulb **212** and/or the light bulb **222** for controlling the respective light bulbs in response to digital messages received from external devices via the network **282**. The hub device **280** may communicate via one or more types of RF communication signals (e.g., ZIGBEE®; NFC; BLUETOOTH®; WI-FI®; cellular; a proprietary communication channel, such as CLEAR CONNECT™, etc.). The hub device **280** may be configured to receive RF signals **206** from the battery-powered remote control device **250**, the light bulb **212**, and/or the light bulb **222** (e.g., using ZIGBEE®; NFC; BLUETOOTH®; or a proprietary communication channel, such as CLEAR CONNECT™, etc.). The hub device **280** may be configured to transmit digital messages via the network **282** for providing data (e.g., status information) to external devices.

The hub device **280** may operate as a central controller for the load control system **200**, and/or relay digital messages between the control devices (e.g., lighting devices) of the load control system and the network **282**. The hub device **280** may be plugged into (e.g., via electrical plug **228**) an electrical receptacle **229** that is powered by the AC power source **202**. The hub device **280** may receive power without being controlled by the same actuator as the light bulbs **212**, **222** (e.g., without being controlled by the toggle actuator **216**). For example, toggle actuator **216** may not turn the hub device **280** on and off. The hub device **280** may receive power from an external power source. For example, the hub device **280** may receive power from a battery. The hub device **280** may be on-site at the load control system **200** or at a remote location. Though the hub device **280** is shown as a single device, the load control system **200** may include multiple hub devices and/or the functionality thereof may be distributed across multiple devices.

The load control system **200** may include a network device **290**, such as, a smart phone (for example, an iPhone® smart phone, an Android® smart phone, or a Blackberry® smart phone), a personal computer, a laptop, a wireless-capable media device (e.g., MP3 player, gaming device, or television), a tablet device, (for example, an iPad® handheld computing device), a Wi-Fi or wireless-communication-capable television, or any other suitable network communication or Internet-Protocol-enabled device. The network device **290** may be operable to transmit digital messages in one or more Internet Protocol packets to the hub device **280** via RF signals **208** either directly or via the network **282**. For example, the network device **290** may transmit the RF signals **208** to the hub device **280** via a Wi-Fi communication link, a Wi-MAX communications link, a Bluetooth® communications link, a near field communication (NFC) link, a cellular communications link, a television white space (TVWS) communication link, or any combination thereof. The RF signals **208** may be communicated using a different protocol and/or wireless band than the RF

signals **206**. For example, the RF signals **208** may be configured for Wi-Fi communication or cellular communication, while RF signals **206** may be configured for ZIGBEE® or a proprietary communication channel, such as CLEAR CONNECT™. In another example, the RF signals **208** and the RF signals **206** may be the same. Examples of load control systems operable to communicate with network devices on a network are described in greater detail in commonly-assigned U.S. Pat. No. 10,271,407, issued Apr. 23, 2019, entitled LOAD CONTROL DEVICE HAVING INTERNET CONNECTIVITY, the entire disclosure of which is hereby incorporated by reference.

The network device **290** may include a visual display **292**. The visual display **292** may include a touch screen that may include, for example, a capacitive touch pad displaced overtop the visual display, such that the visual display may display soft buttons that may be actuated by a user. The network device **290** may include a plurality of hard buttons, e.g., physical buttons (not shown), in addition to the visual display **292**. The network device **290** may download a product control application for allowing a user of the network device **290** to control the load control system **200**. In response to actuations of the displayed soft buttons and/or hard buttons, the network device **290** may transmit digital messages to the light bulb **212**, the light bulb **222**, and/or the hub device **280** through the wireless communications described herein.

The network device **290** may transmit digital messages via the RF signals **208** for controlling the light bulb **212** and/or the light bulb **222**. The network device **290** may transmit digital messages to the light bulb **212** and/or the light bulb **222** via the hub device **280** (e.g., using ZIGBEE®, NFC; BLUETOOTH®, or a proprietary communication channel, such as CLEAR CONNECT™, etc.). For example, the network device **290** may transmit digital messages to turn the light bulbs **212**, **222** on, off, and/or modify the intensities of respective light bulbs **212**, **222**. The network device **290** may vary the intensity of the light bulbs **212**, **222** by varying the amount of power delivered to the light bulbs **212**, **222**. For example, the network device **290** may increase or decrease the intensity of the light bulbs **212**, **222** from a minimum intensity (e.g., approximately 1%) to a maximum intensity (e.g., approximately 100%). The light bulb **212** and/or the light bulb **222** may communicate with the hub device via RF signals **206** (e.g., to transmit status information). For example, the light bulb **212** and/or the light bulb **222** may be configured to communicate with the hub device **280** (e.g., using ZIGBEE®, NFC; BLUETOOTH®, or a proprietary communication channel, such as CLEAR CONNECT™, etc.) in response to digital messages received from the battery-powered remote control device **250** and/or another cause of change in lighting intensity. The hub device **280** may be configured to transmit RF signals **208** to the network device **290**. For example, the RF signals **208** transmitted to the network device **290** may be used for displaying data (e.g., status information) on the visual display **292** of the network device **290**.

The operation of the load control system **200** may be programmed and configured using the hub device **280** and/or the network device **290**. An example of a configuration procedure for a wireless load control system is described in greater detail in commonly-assigned U.S. Pat. No. 10,027,127, issued Jul. 17, 2018, entitled COMMISSIONING LOAD CONTROL SYSTEMS, the entire disclosure of which is hereby incorporated by reference.

The lighting devices (e.g., the light bulb **212**, the light bulb **222**) may include an ability to store information. For

example, the light bulb **212** and/or the light bulb **222** may include memory. The memory may be volatile and/or non-volatile memory. The memory may be used for saving one or more states (e.g., power states) of the lighting devices. For example, the light bulb **212** and/or the light bulb **222** may have an on power state and/or an off power state. The on power state of the light bulb **212** and/or the light bulb **222** may be characterized by an intensity state, a color (e.g., a color temperate) state, etc. The intensity state of the light bulb **212** and/or the light bulb **222** may be the intensity level (e.g., 10%, 50%, 80%) of the light bulb **212** and/or the light bulb **222**. The light bulbs **212**, **222** may be configured to store their respective states in memory. Also, or alternatively, the light bulbs **212**, **222** may be configured to store states of other devices in their respective memory. The light bulbs **212**, **222** may store one or more power states for retention at a later time.

If the hub device **280** is included in the load control system **200**, the hub device **280** may store information. For example, the hub device **280** may include memory that may be volatile and/or non-volatile memory. The memory may be used for saving one or more states (e.g., power states) of the lighting devices and/or one or more states of the hub device **280**. For example, the hub device **280** may have a powered state when receiving power and an unpowered state when insufficient power is being received. The hub device **280** may store the power states of the lighting devices and/or the power states of the hub device **280**. The hub device **280** may store the intensity states and/or color states of the lighting devices. The hub device **280** may store the power states of lighting devices for retention at a later time.

The hub device **280** may be used to set a state (e.g., power state) of one or more of the lighting devices of the load control system **200** (e.g., the light bulb **212**, the light bulb **222**). For example, the hub device **280** may be used to set the state of one or more of the lighting devices to an on power state, an off power state, a defined intensity state (e.g., a preset intensity state), a defined color state, etc. The lighting devices may be in the on power state to be controlled to other states, such as the defined intensity state or the defined color state, for example. The lighting devices may change their respective states to be aligned with the states provided by the hub device **280**. The hub device **280** may change the states of the lighting devices to be aligned with the states stored by the hub device **280** when electrical power is being provided to the lighting devices. For example, the hub device **280** may set a state of the light bulb **212** to an on power state. If power is being provided to the light bulb **212** (e.g., the toggle actuator **216** is turned to the on position), the light bulb **212** may be turned to the on power state. If electrical power is not provided to the lighting load (e.g., a power removal event occurs, such as the toggle actuator **216** is turned to the off position), the hub device **280** may not be able to set the light bulb **212** to the on power state and/or to a defined intensity or color. For example, the hub device **280** may be precluded from setting the state of the light bulb **212** to the on power state or to a defined intensity or color if insufficient electrical power is being provided to the light bulb **212** (e.g., a power removal event occurs, such as the toggle actuator **216** is turned to the off position). The hub device **280** may set a state of the light bulb **212** to the off power state. For example, if power is being provided to the light bulb **212** (e.g., the toggle actuator **216** is turned to the on position), the hub device **280** may set the power state of the light bulb **212** to the off power state, based on the information stored by the hub device **280**. The network device **290** may be used to set

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a state (e.g., power state), or not set a state, of one or more lighting devices in a manner similar to the hub device **280**.

The hub device **280** and/or the network device **290** may set a state (e.g., a power state) of one or more lighting devices by sending a digital message to the light bulbs **112**, **122**. The hub device **280** and/or the network device **290** may send the digital messages directly to the light bulbs **112**, **122**. The hub device **280** and/or the network device **290** may send the digital messages to the light bulbs **112**, **122** via an intermediary device (e.g., via a separate hub device, network device, and/or control device). The digital message may include the power state to which the lighting devices may be set. The light bulbs **112**, **122** may set the respective lighting devices to the received power state. The light bulbs **112**, **122** may store the received power state. The light bulbs **112**, **122** may transmit a digital message to the hub device **280** and/or the network device **290**. For example, the light bulbs **112**, **122** may transmit a digital message to the hub device **280** and/or the network device **290** as an acknowledgement that the light bulbs **112**, **122** have received and/or stored the power state sent by the hub device **280** and/or the network device **290**.

The hub device **280** may monitor when the remote control device **250** changes the power state of the lighting devices and update the power states that are stored in memory. The hub device **280** may identify digital messages from the remote control device **250** that are transmitted to control the power state of the lighting devices. In response to the hub device **280** identifying a command to change the power state in the digital messages transmitted from the remote control device **250**, the hub device **280** may update the power states stored in memory. The hub device **280** may await a confirmation of the change in power state from the lighting devices before updating the stored power state.

A power state as set by the hub device **280** and/or the network device **290** may be different than the actual state of the lighting devices (e.g., light bulb **212**). As described herein, the hub device **280** and/or the network device **290** may be configured to set a power state of the lighting devices (e.g., the light bulb **212**). For example, the hub device **280** and/or the network device **290** may set a power state of the light bulb **212** to an on power state. As long as electrical power is provided to the lighting devices, the lighting devices may set their respective power states to the power states provided by the hub device **280** and/or the network device **290**. The power states (e.g., on power states) may be overridden by a power removal event during which the actual states of the lighting devices may be different than the power states stored by the hub device **280** and/or the network device **290**. For example, if power is removed from the lighting load via a power removal event, the light bulb **212** may be in an off power state. A power removal event may include a user turning the toggle actuator **216** to the off position. For example, if a user turns the toggle actuator **216** to the off position, the light bulb **212** may be turned to an off power state, even if the hub device **280**, the network device **290**, and/or the remote control device **250** has set the power state of the light bulb **212** to the on power state.

A power removal event may also include a power outage (e.g., a blackout and/or a brownout). For example, power may be removed from the lighting devices if one or more portions (e.g., rooms) of a load control system **200** experiences a blackout, brownout, etc. If power is removed from the light bulb **212** (e.g., via a power removal event, such as a blackout), the light bulb **212** may be turned to an off power state, notwithstanding that the hub device **280**, the network

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device **290**, and/or remote control device **250** has set the power state of the light bulb **212** to be an on power state.

The prior power states (e.g., the power states of lighting devices set prior to a power removal event) may be stored by the light bulb **212**, the light bulb **222**, the hub device **280**, the network device **290**, and/or stored in an external database (e.g., cloud database) via the network **282**. The prior power states of lighting devices may be recalled at a predefined time. The prior power states of the lighting devices may be recalled after a predetermined event (e.g., upon a user action).

A power removal event may be a local power removal event or a system power removal event. A local power removal event may occur as a result of a user controlling the toggle actuator **216** and/or the actuator **225** to an off position. For example, a user may actuate the toggle actuator **216** and/or the actuator **225** in an attempt to turn off the power and/or functionality of the respective light bulbs **212**, **222**. A single control device or multiple control devices located within a location (e.g., one or more control devices located within a room or controlled by the same switch) may experience a local power removal event. A system power removal event may be a blackout, brownout, or other power removal event that affects the devices in the load control system **200**. The hub device **280** may be able to determine whether a power removal event is a system power removal event (e.g., a blackout, brownout, etc.). A system power removal event may involve the removal of power to the hub device **280** and one or more control devices within the load control system **200**. In response to the system power removal event, the hub device **280** may operate in an unpowered state (e.g., powered off). The hub device **280** may enter a battery powered mode after detecting the loss of power, or may remain off until the return of power from the AC circuit. The hub device **280** may detect a return to the powered state and may subsequently control the light bulbs **212**, **222** to a prior power state (e.g., prior to the loss of power). The light bulbs **212**, **222** may also, or alternatively, store their prior power states locally and recall the prior power states after a system power removal event is detected (e.g., via an indication from the hub device **280**).

After the lighting devices (e.g., light bulbs **212**, **222**) and/or the hub device **280** determines that the power removal event is a local power removal event at one of the lighting devices (e.g., the user turning off power to an AC circuit), the lighting devices and/or the hub device **280** may refrain from transmitting a prior power state. After a local power removal event has ended and power is returned to the lighting devices affected by the local power removal event (e.g., the user turning on the power to the AC circuit), lighting devices affected by the local power removal event may turn on to a default intensity level (e.g., maximum intensity level, a defined intensity level, such as eighty-five percent, etc.) or to a prior intensity level. The hub device **280** and/or other lighting devices may refrain from transmitting messages to detect whether a power removal event has occurred and/or from controlling the power state of the affected lighting devices. For example, after the light bulb **222** and/or the hub device **280** determines that a power removal event to the light bulb **212** was a local power removal event (e.g., because toggle actuator **216** was actuated to the off position), the light bulb **222** and/or the hub device **280** may refrain from transmitting a digital message to control the power state of the light bulb **212**.

After the lighting devices and/or the hub device **280** determines that the power removal event is a system power removal event (e.g., a brownout, blackout, etc.), the lighting

devices and/or the hub device **280** may modify the power states of the lighting devices to the prior power states of the lighting devices. For example, upon the light bulbs **212**, **222** and/or the hub device **280** determining that the power removal event to the light bulbs **212**, **222** is a system power removal event (e.g., by determining that the light bulb **212** and the light bulb **222** have experienced a power removal event), the light bulbs **212**, **222** and/or the hub device **280** may set the power state of the lighting devices to the respective prior power states. Thus, in the event of a system power removal event, the light bulbs **212**, **222** may be prevented from operating at an undesirable default intensity level (e.g., maximum intensity level, a defined intensity level, such as eighty-five percent, etc.) when power is returned to the light bulbs **212**, **222**, and may even remain in an off power state when that is the prior power state of the light bulbs **212**, **222** before the system power removal event.

In an example, the light bulb **212** may be in the off power state (e.g., zero percent intensity level), but still receiving power from the AC power source **202**. If the light bulb **212** experiences a local power removal event, and power is returned to the light bulb **212**, the light bulb **212** may turn on to a default intensity level, such as a maximum intensity level or a defined intensity level, e.g., eighty-five percent. If the light bulb **212** experiences a system power removal event, and the hub device **280** is implemented in the system **200**, the hub device **280** may detect the system power removal event and send a digital message that overrides the default intensity level of the light bulb **212** with the stored prior power state (e.g., zero percent intensity level) of the light bulb **212** before the system power removal event. If the light bulb **212** stores the prior power state (e.g., zero percent intensity level) of the light bulb **212** locally and detects that a system power removal event has occurred, the light bulb **212** may override the default power state with the prior power state of the light bulb **212** before the system power removal event and remain in the off power state (e.g., zero percent intensity level). In the examples provided, the light bulb **212** may remain off, or return to another prior power state, in the event of a system power removal event.

The lighting load of the light bulb **212** may operate at the recalled prior power state at a predefined time and/or the lighting load of the light bulb **212** may function according to the recalled prior power state after a predefined duration (e.g., five minutes after the power is returned to the lighting load and/or immediately after the power is returned to the lighting load). For example, upon a system power removal event beginning and ending, the light bulb **212** may recall (e.g., from memory) that the prior power state of the lighting load was an on power state at an intensity state of eighty percent. Upon the power removal event ending, the light bulb **212** may operate the lighting load in the on power state at the intensity state of eighty percent. The lighting load may function at eighty percent in the morning, and/or the lighting load may function at the defined intensity upon a predefined duration (e.g., ten minutes, immediately, etc.) after the power removal event ends.

Other devices (e.g., the hub device **280** and/or the network device **290**) within the load control system **200** may be used to coordinate the storage and/or retrieval of power states of lighting devices. The hub device **280** and/or the network device **290** may be used to store power states of lighting devices prior to a power removal event (e.g., as the prior power states). The hub device **280** and/or the network device **290** may coordinate sending messages, such as messages including power states, among lighting devices. A button (e.g., a "soft" button) on the network device **290** may be

actuated to set lighting devices to predefined power states. For example, a button on the network device **290** may be actuated to set the light bulb **212** to a power state of forty percent intensity level. Upon the conclusion of a power removal event, the light bulb **212** may be set to the power state set by the network device **290** (e.g., an intensity level of forty percent). The light bulb **212** may retrieve the intensity level of forty percent internally (e.g., from memory within the light bulb **212**) and/or the light bulb **212** may retrieve the intensity level of forty percent externally (e.g., from the hub device **280**, the network device **290** and/or one or more other lighting devices).

The prior power state may be pushed to the light bulb light bulbs **212**, **222** after the identification of the end of the power removal event without the light bulbs **212**, **222** sending a request. For example, the hub device **280** and/or the network device **290** may send a digital message to the light bulb **212** that indicates a prior power state to which the light bulbs **212**, **222** may each be controlled after the hub device **280** and/or the network device **290** detect the end of a power removal event (e.g., by detecting a change from a powered state to an unpowered state, and back to a powered state).

The hub device **280** and/or the network device **290** may detect the occurrence of a power removal event (e.g., by detecting a change from a powered state to an unpowered state, and back to a powered state) and query the light bulbs **212**, **222** to determine whether the light bulbs **212**, **222** also experienced the power removal event. The query may include a time or timeframe within which the power removal event was detected at the hub device **280** and/or the network device **290**. The time may be a time at which the power removal event began and/or the time at which the power removal event ended. The query may include a device identifier (e.g., unique identifier) of the devices that are intended to respond to the query. The light bulbs **212**, **222** may acknowledge whether a power removal event was experienced by transmitting an acknowledgement message to the hub device **280** when the light bulbs **212**, **222** experienced the power removal event. The acknowledgement message may include the time and/or timeframe in which the power removal event was detected at the respective light bulbs **212**, **222**, such that the hub device **280** and/or the network device **290** may determine whether the power removal events coincide between the devices. The hub device **280** and/or the network device **290** may determine that the power removal event is a system power removal event when the queried light bulbs **212**, **222** also experienced the power removal event and send the prior power state to the light bulbs **212**, **222**. The hub device **280** and/or the network device **290** may send to and/or query the light bulbs **212**, **222** periodically, at a time of day, and/or aperiodically (e.g., based on a user action).

The light bulbs **212**, **222** may transmit a digital message to the hub device **280** that indicates the occurrence of a power removal event without being requested. The digital messages sent from the light bulbs **212**, **222** may indicate the power states of the light bulbs **212**, **222**. The digital messages may indicate whether a power removal event has occurred to one or more other lighting devices. The digital message may identify the lighting devices by a name, a number (e.g., unique identifier), etc. The light bulbs **212**, **222** may send to and/or query the hub device **280** and/or the network device **290** periodically, at a time of day, and/or aperiodically (e.g., based on a user action).

The light bulbs **212**, **222** may query for a prior power state after detection of a power removal event. For example, the hub device **280** and/or the network device **290** may send a

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digital message to the light bulb **212** in response to a request for a prior power state by the light bulbs **212**, **222** after the occurrence of a power removal event. The light bulbs **212**, **222** may send a digital message that indicates that a power removal event occurred, which may be interpreted by the hub device **280** and/or the network device **290** as a request for a prior power state. The digital message may include a time or timeframe associated with the power removal event. If the hub device **280** and/or the network device **290** did not experience the power removal event (e.g., within a coinciding time period), the hub device **280** and/or the network device **290** may prevent the sending of the prior power state.

The light bulbs **212**, **222** may send a message to the other devices in the system **200**. For example, the light bulb **212** may send a message to the other devices that the light bulb **212** experienced a power removal event, which may indicate that the light bulb **212** has regained power after the power removal event, and the other devices in the system **200** (e.g., the hub device **280**, the network device **290** and/or one or more other lighting devices) may send a digital message indicating a power state at which the light bulb **212** should operate.

Though the messages indicating power states and/or identification of power removal events may be described herein as being performed by the hub device **280** and/or the network device **290**, such functionality may be implemented at the remote control device **250**, and/or at the control devices (e.g., lighting devices) themselves. For example, one or more lighting devices, such as the light bulb **212**, the light bulb **222**, etc., may be used to store the prior power states, identify the occurrence of a power removal event (e.g., locally and/or by querying other devices), and/or communicate the prior power states to one or more other lighting devices in response to a system power removal event. The digital messages indicating the power states and/or querying for acknowledgement of a power removal event may be received by lighting devices neighboring the lighting devices sending the messages and/or within a similar location as the lighting devices sending the messages. The lighting devices receiving the messages may set their power state to the power state included in the digital messages, or respond to the requests for acknowledgement of a power removal event when such a power removal event has occurred locally.

The lighting devices and/or the hub device **280** may be configured according to one or more locations (e.g., rooms). The lighting devices and/or the hub device **280** may be configured according to one or more sources of power (e.g., AC power source **202**). The lighting devices and/or the hub device **280** may be configured according to one or more AC circuits. A lighting devices and/or the hub **280** may be identified by a corresponding location, AC power source, and/or AC circuit. The location, AC power source, and/or AC circuit corresponding to a lighting device and/or the hub device **280** may be stored by the lighting devices and/or the hub device **280**.

The lighting devices and/or the hub device **280** may receive power from one or more AC circuits. For example, the light bulb **212** may receive power from AC power source **202**. The light bulb **212** may receive power from AC power source **202** via an AC circuit. The toggle actuator **216** may be actuated so that power is provided to the light bulb **212** or so that power is not provided to the light bulb **212**. The light bulb **222** and the hub device **280** may receive power from AC power source **202**. The light bulb **212** may receive power from AC power source **202** via an AC circuit that is different than the AC circuit providing power to light bulb

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212. For example, the light bulb **222** may receive power from the electrical receptacle **226** and/or the hub device **280** may receive power from electrical receptacle **229**. The electrical receptacles **226**, **229** may not be controlled by the actuation of a switch (e.g., toggle actuator **216**). The lighting devices and/or the hub device **280** may identify light bulb **212** as being controllable by a switch (e.g., toggle actuator **216**). The lighting devices and/or the hub device **280** may identify light bulb **222** as not being controllable by a switch (e.g., toggle actuator **216**).

The lighting devices and/or the hub device **280** may determine whether power is provided to one or more lighting devices and/or hub devices. The lighting devices and/or the hub device **280** may determine whether power is provided to one or more lighting devices and/or hub devices according to the AC circuits from which the lighting devices and/or the hub device **280** receive electrical power. For example, the lighting devices and/or the hub device **280** may determine if electrical power is provided to the light bulb **212**. The lighting devices and/or the hub device **280** may also, or alternatively, determine if electrical power is provided to the light bulb **222** and/or the hub device **280**. The lighting devices and/or the hub device **280** may determine the type of power removal event (e.g., local power removal event, system power removal event) by determining if one or more AC circuits have been affected by a power removal event. For example, the hub device **280** may determine that a power removal event is a system power removal event (e.g., a brownout and/or a blackout) by determining that multiple AC circuits (e.g., AC circuits providing electrical power to the light bulb **212**, the light bulb **222**, and/or the hub device **280**) have experienced a power removal event. The lighting devices and/or the hub device **280** may query devices (e.g., lighting devices, the hub device **280**, and/or other control devices) connected to different AC circuits to determine if the power removal event is a system power removal event (e.g., a power removal event affecting more than one AC circuit, such as a brownout or blackout) and/or whether the power removal event is a local power removal event (e.g., a power removal event affecting a single AC circuit), such as when a user physically turns off the toggle actuator **216** that controls the light bulb **212**. The lighting devices and/or the hub device **280** may query devices positioned within different locations (e.g., rooms) to determine if a power removal event is a system power removal event or a local power removal event.

Setting the power state of the lighting devices to the respective prior power states may be controlled by the user. For example, a user may set the power state of the light bulb **212** to fifteen percent (e.g., when the user is away on vacation). If a system power removal event begins and ends, the user may apply a setting that is stored at the hub device **280** and/or the lighting device **212** to automatically control the light bulb **212** to an intensity level of fifteen percent when power is again provided to the light bulb **212** (e.g., when the power removal event ends). As another example, a user may set the intensity level of the light bulb **212** to zero percent (e.g., prior to the user being away or going to bed). If a system power removal event occurs while the user is away or is in bed, the intensity level that was previously set by the user may be used to control the light bulb **212** to an intensity of zero percent when power is again provided to the light bulb **212** (e.g., when the power removal event ends). The user settings may be stored and applied similar to the prior power state that is stored and applied herein. User settings may be given priority over a prior power state that has been stored after the user settings, or the prior power

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state may be given priority over the user settings. Although the control devices (e.g., lighting devices) may be configured to provide light at an intensity displayed prior to a system power removal event, this may be disabled by the user. For example, a user may configure the control devices so that they provide light at a default intensity level (e.g., 100% intensity level, 85% intensity level, etc.) upon a system power removal event ending.

The hub device **280** and/or the lighting devices may determine whether a power removal event is a local power removal event or a system power removal event using one or more factors. For example, the hub device **280** and/or the lighting devices may determine that a power removal event is a local power removal event or a system power removal event based on the number of control devices (e.g., lighting devices) that have experienced a power removal event. The hub device **280** and/or the lighting devices may set a threshold for the number of devices that experienced the power removal event indicating that the power removal event was a system power removal event. If each of the lighting devices within a load control system have experienced a power removal event, it may be determined that the power removal event was a system power removal event. If one or more lighting devices within a load control system have not experienced a power removal event at the same time, power removal events experienced by the control devices may be determined to be local power removal events.

The hub device **280** and/or the lighting devices may determine that a power removal event is a local power removal event or a system power removal event based on the type of the control devices that have experienced a power removal event. Some control devices may not be downstream of a switch (e.g., the wall-mounted load control device **210**). For example, a dimmer device may not be downstream of a switch and may be configured to control the power delivered to a lighting device, such as a dimmable light source. Other devices may be downstream of a switch. The light bulb **212**, shown in FIG. 2A, may be an example of a device that is downstream of a switch.

A weighting factor may be used for determining whether a power removal event is a local power removal event or a system power removal event. For example, control devices (e.g., lighting devices) not downstream of a switch may be provided a higher weight than control devices downstream of a switch, when determining whether a power removal event was a local power removal event or a system power removal event. For example, a dimmer device (such as dimmer device **218**, shown in FIG. 2B) experiencing a power removal event may provide a larger weight than the light bulb **212** experiencing a power removal event, when determining whether a power removal event was a local power removal event or a system power removal event. If the light bulb **212** experiences a power removal event and the dimmer device continues receiving power at the time of the power removal event at the bulb **212**, it may be determined that the power removal event experienced by the light bulb **212** is a local power removal event. If dimmer device experiences a power removal event, the hub device **280** and/or the lighting devices may determine that the power removal event experienced by the dimmer device is a system power removal event.

The light bulb **212**, the light bulb **222**, and the hub device **280** may be electrically coupled to AC power source **202**. Although the light bulb **212**, the light bulb **222**, and the hub device **280** are electrically coupled to the same power source (e.g., AC power source **202**), the light bulb **212** is on a

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different circuit than the light bulb **222** and the hub device **280**. The light bulb **212** is electrically coupled to the wall-mounted load control device **210**. The light bulb **222** and the hub device **280** are not electrically coupled to the wall-mounted load control device **210**. The hub device **280** and/or lighting devices may query more than one control device (e.g., the light bulb **212** and the light bulb **222**) to determine if a power removal event is a system power removal event or a local power removal event. The hub device **280** and/or lighting devices may query the hub device **280** and one or more control devices to determine if a power removal event is a system power removal event or a local power removal event. For example, the hub device **280** may query the light bulb **212** and the light bulb **222** to determine if a power removal event is a system power removal event or a local power removal event. If the light bulb **212** experienced a power removal event and the light bulb **222** experienced a power removal event, the hub device **280** may determine that the power removal events are system power removal events (e.g., because more than one control device on more than one AC circuit experienced a power removal event). If the light bulb **212** experienced a power removal event, the light bulb **222** experienced a power removal event, and the hub device **280** experienced a power removal event, it may further be determined that the power removal events are system power removal events. If the light bulb **212** experienced a power removal event and the light bulb **222** did not experience a power removal event, the hub device **280** may determine that the power removal event to the light bulb **212** was a local power removal event (e.g., the light bulb **212** experienced a power removal event when a user turned the toggle actuator **216** to the off position).

The hub device **280** may query control devices and/or hub devices in one or more locations (e.g., rooms) to determine if a power removal event was a system power removal event or a local power removal event. For example, if power removal events occurred within more than one room, the hub device **280** may determine that the power removal event was a system power removal event. The hub device **280** may query, on a periodic basis (e.g., every 10 seconds, 5 minutes, 1 hour, etc.) whether the control devices have experienced a power removal event. The hub device **280** may query, aperiodically (e.g., based on a user actuation at the hub device **280**), whether the control devices have experienced a power removal event. Also, or alternatively, the hub device **280** may query control devices to determine if the control devices have experienced a power removal upon the hub device **280** regaining power after experiencing a power removal event.

The control devices (e.g., lighting devices, such as the light bulb **212**) may query control devices (e.g., other lighting devices, such as the light bulb **222**) to determine whether the control devices have experienced a power removal event. The light bulb **212** may determine whether the light bulb **212** has experienced a power event. The light bulb **212** may determine whether the power removal event of the other control device (e.g., the light bulb **222**) and/or the power removal event of the light bulb **212** was a system power removal event or a local power removal event. For example, the light bulb **212** may query the light bulb **222** to determine whether the light bulb **222** has experienced a power removal event. The light bulb **212** may query the light bulb **222** and/or access local storage to determine if the light bulb **222** and/or the light bulb **212** has experienced a power removal event.

The hub device **280** and/or control devices (e.g., lighting devices) may perform a query to determine whether a power

removal event occurred, and/or the type of the power removal event, based on one or more conditions. For example, the lighting devices (e.g., the light bulb **212**) may perform a query to determine whether a power removal event occurred, and/or the type of the power removal event, after the light bulb **212** loses and regains power. The light bulb **212** may query other lighting devices after regaining power and/or the light bulb **212** may query the hub device **280** after regaining power. After regaining power, the light bulb **212** may query other lighting devices and/or the hub device **280** to determine whether the other lighting devices and/or the hub device **280** have experienced, and/or are experiencing, a power removal event.

If the other lighting devices and/or the hub device **280** have experienced, and/or are experiencing, a power removal event, the light bulb **212** may determine whether the power removal event experienced by the other control devices and/or the hub device **280** occurred at the same time that the light bulb **212** lost power. The light bulb **212** may perform this determination by requesting the time of the power removal events, the duration of the power removal events, etc. For example, if the other control devices and/or the hub device **280** experienced a power removal event at the same time, and/or for the same duration that the light bulb **212** experienced the power removal event, the power removal event may be determined to be a system power removal event. If the other control devices and/or the hub device **280** did not experience a power removal event, and/or if the other control devices and/or the hub device **280** did not experience a power removal event at the same time as the light bulb **212**, the power removal event may be determined to be a local power removal event.

The hub device **280** may perform a query to determine whether a power removal event occurred, and/or the type of the power removal event, after the hub device **280** loses and regains power. The hub device **280** may query lighting devices after regaining power. The hub device **280** may query other hub devices after regaining power. After regaining power, the hub device **280** may query other hub devices and/or lighting devices to determine whether the other hub devices and/or lighting devices have experienced, and/or are experiencing, a power removal event.

If the other hub devices and/or the lighting devices have experienced, and/or are experiencing, a power removal event, the hub device **280** may determine whether the power removal event experienced by the other hub devices and/or the lighting devices occurred at the same time that the hub device **280** lost power. The hub device **280** may perform this determination by requesting the time of the power removal events, the duration of the power removal events, etc. For example, if the other hub devices and/or the lighting devices experienced a power removal event at the same time, and/or for the same duration, that the hub device **280** experienced the power removal event, the power removal event may be determined to be a system power removal event. If the other hub devices and/or the lighting devices did not experience a power removal event, and/or if the other hub devices and/or the lighting devices did not experience a power removal event at the same time as the hub device **280**, the power removal event may be determined to be a local power removal event.

The hub device **280** and/or the lighting devices may perform a query to determine whether a power removal event occurred, and/or the type of the power removal event, periodically. For example, the hub device **280** and/or the lighting devices may perform a query to determine whether a power removal event occurred, and/or the type of the

power removal event at intervals of five minutes, thirty minutes, two hours, etc. The hub device **280** and/or the lighting devices may perform a query to determine whether a power removal event occurred, and/or the type of the power removal event, aperiodically. For example, the hub device **280** and/or the lighting devices may perform a query to determine whether a power removal event occurred, and/or the type of the power removal event, based on a user action and/or based on a time of day. The hub device **280** and/or the lighting devices may perform a query to determine whether a power removal event occurred, and/or the type of the power removal event, based on one or more of the factors, or a combination of the factors, described herein.

The hub device **280** and/or the lighting devices may perform a query to determine whether a power removal event occurred, and/or the type of the power removal event, based on receiving a message from one or more other devices. For example, another control device may send the hub device **280** and/or lighting devices a message. The message may request that the hub device **280** and/or the lighting devices query other control devices and/or hub devices to determine whether a power removal event occurred, and/or the type of the power removal event that may have occurred. The other control devices may send this message to the hub device **280** and/or the lighting devices for one or more reasons. For example, the other control devices may send this message to the hub device **280** and/or the lighting devices after the control devices regain power (e.g., the other control devices regain power after a power removal event). The other control devices may send this message to the hub device **280** and/or the lighting devices, e.g., periodically or aperiodically.

As described above, the light bulb **212**, the light bulb **222**, and the hub device **280** are coupled to AC power source **202**. Although the light bulb **212**, the light bulb **222**, and the hub device **280** are coupled to the same power source (e.g., AC power source **202**), the light bulb **212** is on a different circuit than the light bulb **222** and the hub device **280**. The light bulb **212** is coupled to the wall-mounted load control device **210**. The light bulb **222** and the hub device **280** are not coupled to the wall-mounted load control device **210**. The light bulb **212** may query one or more control devices to determine if a power removal event is a system power removal event or a local power removal event. The light bulb **212** may query the hub device **280** and one or more control devices to determine if a power removal event is a system power removal event or a local power removal event. For example, the light bulb **212** may query the light bulb **222** to determine if a detected power removal event is a system power removal event or a local power removal event. If the light bulb **222** also experienced a power removal event, the light bulb **212** may determine that the power removal events are system power removal events (e.g., because more than one control device on more than one AC circuit experienced a power removal event). If the light bulb **212** experienced a power removal event, the light bulb **222** experienced a power removal event, and the hub device **280** experienced a power removal event, it may further be determined that the power removal events are system power removal events. If the light bulb **212** experienced a power removal event and the light bulb **222** did not experience a power removal event, the light bulb **212** may determine that the power removal event to the light bulb **212** was a local power removal event (e.g., the light bulb **212** experienced a power removal event when a user turned the toggle actuator **216** to the off position).

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The light bulb **212** may query other lighting devices and/or the hub devices in one or more locations (e.g., rooms) to determine if a power removal event was a system power removal event or a local power removal event. For example, if power removal events occurred within more than one room, the light bulb **212** may determine that the power removal event was a system power removal event. The light bulb **212** may query, on a periodic basis (e.g., every 10 seconds, 5 minutes, 1 hour, etc.) whether other lighting devices and/or the hub devices have experienced a power removal event. The light bulb **212** may query, aperiodically (e.g., based on a user action), whether the other lighting devices and/or the hub devices have experienced a power removal event. Also, or alternatively, the light bulb **212** may query the other lighting devices and/or the hub devices to determine if the other lighting devices and/or the hub devices have experienced a power removal upon the light bulb **212** regaining power after experiencing a power removal event.

The hub device **280** may comprise an additional source of power (e.g., an internal or external battery). The battery may provide power to the hub device **280** in the event that electrical power from an AC power source (such as AC power source **202**, shown in FIG. 2) ceases to provide power. For example, the hub device **280** may receive electrical power from a battery if the hub device **280** experiences a power removal event (e.g., a system power removal event and/or a local power removal event). The hub device **280** may be configured to determine that it is not receiving power from the AC power source, but is being powered from the battery. If the hub device **280** experiences a power removal event and the hub device **280** begins receiving power from the battery, the hub device **280** may query control devices (e.g., lighting devices, such as the light bulb **212** and the light bulb **222**) to determine if the power removal event to the hub device was a system power removal event or a local power removal event.

For example, after the hub device **280** experiences a power removal event and receives power from a battery, the hub device **280** may query the light bulb **212** to determine if light bulb **212** has experienced a power removal event. The hub device **280** may query the light bulb **212** to determine whether the light bulb experienced a power removal event at the same time as the hub device **280**. If the light bulb **212** experienced a power removal event, and/or if the light bulb **212** experienced the power removal event at the same time as the hub device, the hub device **280** may determine that the power removal event was a system power removal event. If the light bulb **212** did not experience a power removal event, and/or if the light bulb **212** experienced a power removal event at a different time than the hub device **280**, the hub device **280** may determine that the power removal event of the hub device was a local power removal event. The hub device may query one or more other control devices (e.g., the light bulb **222**) to determine if the power removal event experienced by the hub device was a system power removal event or a local power removal event.

FIG. 2B shows a load control system **200a** having a wall-mounted load control device (e.g., a dimmer device **218**) for controlling the amount of power delivered to a lighting device, such as a light bulb **212a** (e.g., a lighting load). As shown in FIG. 2B, the light bulb **212a** may comprise a standard, dimmable light bulb (e.g., a light bulb that is not responsive to wireless signals). The dimmer device **218** may be configured to control the amount of power delivered to the light bulb **212a** using a phase-control dimming technique. The dimmer device **218** may include

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actuators, such as buttons **219**. The dimmer device **218** may be configured to control the power delivered to the light bulb **212a** in response to actuation of buttons **219**. The dimmer device **218** may be configured to receive digital messages via wireless signals, e.g., radio-frequency (RF) signals **206** (e.g., ZIGBEE®, NFC; BLUETOOTH®; WI-FI®; or a proprietary communication channel, such as CLEAR CONNECT™, etc.). Examples of wall-mounted dimmer devices are described in greater detail in U.S. Pat. No. 5,248,919, issued Sep. 28, 1993, entitled LIGHTING CONTROL DEVICE, and U.S. Pat. No. 9,679,969, issued Jun. 13, 2017, entitled WIRELESS LOAD CONTROL DEVICE, the entire disclosures of which are hereby incorporated by reference.

The dimmer device **218** may be configured to control the power delivered from the AC power source **202** to the light bulb **212a** to control the intensity level of the light bulb **212a**. The dimmer device **218** may retain the power state of the light bulb **212a**. The dimmer device **218** may be configured to retain the power state of the light bulb **212a** to an on power state. For example, dimmer device **218** may be configured to retain an intensity level of the light bulb **212a** to a non-zero value (e.g., an intensity level of 1%-100%). The dimmer device **218** may retain the power state of the light bulb **212a** to an on power state, for example, by physically maintaining electrical continuity of the circuit (e.g., AC circuit) between AC power source **202** and the light bulb **212a**. The dimmer device **218** may receive the power from the AC power source **202** and prevent the power from being sent to the light bulb **212a** to turn the light bulb **212a** to the off power state. The dimmer device **218** may, however, store a time at which the power is received **202** and/or lost from the AC power source **202**. The dimmer device **218** may identify when a power outage is a system power outage, as the dimmer device **218** may receive power so long as the AC power source **202** is providing power. Because the dimmer device **218** may receive power so long as the AC power source **202** is providing power (e.g., the dimmer device **218** is not downstream of a switch, such as wall-mounted load control device **210**), the dimmer device **218** may be considered to be an unswitched electrical device. The load control system **200a** may also comprise the light bulb **212**, as shown in FIG. 2A, that receives power from the AC power source **202** via the wall-mounted load control device **210** and is responsive to the RF signals **206**.

Referring again to FIG. 2B, as the dimmer device **218** may receive power from the AC power source **202** so long as the AC power source **202** is providing power, the light bulbs **212a**, **222** and/or the hub device **280** may query the dimmer device **218** to determine whether a power removal event was a local power removal event or a system power removal event. Because the dimmer device **218** may receive power from the AC power source **202** so long as the AC power source **202** is providing power, it may be more reliable for light bulbs **212a**, **222** and/or the hub device **280** to query the dimmer device **218** to determine the type of power removal event than querying other devices (e.g., light bulbs **212a**, **222** and/or hub device **280**), which may be unplugged from the power source and/or turned off by a light switch. The dimmer device **218** may be an indicator that the power removal event was a system power removal event. If there are multiple dimmer devices in the load control system **200**, each of the dimmer devices losing power may indicate a system power removal event, while some or none of the dimmer devices losing power may indicate a local power removal event. The indication of the dimmer devices losing power may be used in combination to with other devices (e.g., the light bulbs **212a**, **222**, the hub device **280**, etc.) to

determine whether a power removal event was a local power removal event or a system power removal event.

The dimmer device **218** may not experience a local power removal event. For example, because the dimmer device **218** does not receive power from the AC power source **202** through a switch (e.g., the wall-mounted load control device **210**), the dimmer device **218** may not be turned off by a user. A dimmer device **218** may experience a system power removal event. For example, a dimmer device **218** may experience a blackout. Because the dimmer device **218** may not experience a local power removal event, a power removal event experienced by the dimmer device **218** may be determined to be a system power removal event. The hub device **280** and/or lighting devices (e.g., the light bulb **212a** and the light bulb **222**) may determine that the power removal event experienced by the dimmer device **218** is a system power removal event.

As another example, the light bulb **222** may experience a power removal event. The light bulb **222** receives electrical power from electrical receptacle **226**, which is not coupled to dimmer device **218**. Because the light bulb **222** is not coupled to dimmer device **218**, the power removal event to the light bulb **222** may be determined to be a local power removal event (e.g., in response to an actuation of the actuator **225** of the table lamp **224**) and/or the power removal event to the light bulb **222** may be determined to be a system power removal event. To determine whether a power removal event is a local power removal event or a system power removal event, the power states of one or more lighting devices, dimmer devices, and/or hub devices may be determined. For example, if the light bulb **222** is experiencing a power removal event and the dimmer device **218** is experiencing a power removal event, it may be determined that the power removal event is a system power removal event. If the light bulb **222** is experiencing a power removal event and the dimmer devices positioned within one or more rooms are experiencing a power removal event, it may be determined that the power removal event is a system power removal event. If the light bulb **222** is experiencing a power removal event and the dimmer device **218** is not experiencing a power removal event, and vice-versa, it may be determined that the power removal event is a local power removal event. The lighting devices and/or the hub device **280** may determine the power states of one or more lighting devices.

After the lighting devices and/or other devices (e.g., the hub device **280**) determine that the power removal event is a system power removal event, the power states of the lighting devices may be adjusted upon the power removal event ending. For example, upon the power removal event ending, the lighting devices and/or other the hub device **280** may adjust the power state of the lighting devices to prior power states of the lighting devices (e.g., power states prior to the power removal event). For example, after the lighting devices and/or the hub device **280** determine that the power removal event is a system power removal event, the lighting devices and/or the hub device **280** may obtain the prior power states of the lighting devices (e.g., via memory of the lighting devices, memory of the hub device **280**, memory of the network device **290**, memory of an external server, etc.). The lighting devices and/or the hub device **280** may send and/or set the power states of the lighting devices to the obtained prior power states of the lighting devices.

FIG. 2C shows a load control system **200b** having a remote control device **270** that may be mounted over a toggle actuator **216** of the wall-mounted load control device **210** (e.g., such as the toggle actuator **216** shown in FIG. 2A).

The lighting devices (e.g., light bulbs **212**, **222**) may be turned on or off, or adjust the intensity level, in response to digital messages transmitted by the remote control device **270** via the RF signals **206**. For example, the light bulbs **212**, **222** may be turned on or off by turning the remote control device **270**, or pressing the remote control device **270**. The intensity level of the light bulbs **212**, **222** may be increased or decreased by rotating the remote control device **270** in one direction or another, respectively.

The remote control device **270** may be a retrofit remote control device capable of being mounted over the toggle actuator **216** of the wall-mounted load control device **210**. The remote control device **270** may be coupled to the toggle actuator **216** to retain a power state of the lighting devices controlled by the wall-mounted load control device **210**. For example, the remote control device **270** may be configured to maintain the switch installed on the wall-mounted load control device **210** in the “on” position (e.g., by covering the switch when in the “on” position) to maintain the flow of power from the AC power source **202** to the lighting bulb **212**. Because the remote control device **270** may be configured to maintain the switch installed on the wall-mounted load control device **210** in the “on” position (e.g., the light bulb **212** is not downstream of a switch that is able to be controlled to an “off” position), the light bulb **212** downstream of the wall-mounted load control device **210** with the remote control device **270** mounted thereto may be considered to be an unswitched electrical device. The remote control device **270** may be configured to attach to the toggle actuator **216** and hold toggle actuator **216** in the “on” position. Because toggle actuator **216** is maintained in an “on” position, the light bulb **212** may be held in an on power state.

The remote control device **270** may communicate with other devices (e.g., light bulbs **212**, **222**, the network device **290**, and/or the hub device **280**) to transition the power states of light bulbs **212**, **222**. For example, the remote control device **270** may communicate digital messages via the RF signals **206** with light bulbs **212**, **222** and/or the hub device **280** to transition the power states of light bulbs **212**, **222**. The remote control device **270** may transmit digital messages including a move-to-level command that identifies a lighting level to which the lighting devices may change. The move-to-level command may indicate an “on” event or an “off” event to turn the light bulbs **212**, **222** on or off, respectively. For example, the “on” event may be indicated with a 100% lighting level, or another defined lighting level. The “off” event may be indicated with a 0% dimming level. The lighting level for the “on” event and/or the “off” event may also, or alternatively, be stored at the light bulbs **212**, **222** and the light bulbs **212**, **222** may change to the lighting level upon receiving an indication of the occurrence of the “on” event or “off” event from the remote control device **270**. The digital messages may indicate an “on” event when the remote control device **270** is rotated a predefined distance in one direction. The digital messages may indicate an “off” event when the remote control device **270** is rotated a predefined distance in the opposite direction. The digital messages may indicate an “on” event or an “off” event when the remote control device **270** is pressed (e.g., when a button on the face of the remote control device is pressed or the remote control device **270** is pressed in).

The remote control device **270** may transmit digital messages configured to increase the lighting level of the light bulbs **212**, **222** when the remote control device **270** is rotated in one direction. The remote control device **270** may transmit digital messages configured to decrease the lighting

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level of the light bulbs **212**, **222** when the remote control device **270** is rotated in the opposite direction. The digital messages may include a move-with-rate command, which may cause the light bulbs **212**, **222** to change their respective intensity level by a predefined amount. The move-with-rate command may cause the light bulbs **212**, **222** to retain their proportional intensity levels, and/or difference in respective intensity levels. The remote control device **270** may send digital messages to increase or decrease the lighting level by a predefined amount when rotated a predefined distance. The amount of the increase or decrease may be indicated in the digital messages or may be predefined at the light bulbs **212**, **222**.

The lighting devices (e.g., the light bulbs **212**, **222**) and/or the hub device **280** may be associated with the remote control device **270**. The unique identifier (e.g., the serial number) of the remote control device **270** may be stored at the associated devices for recognizing digital messages therefrom for operating according to instructions in the digital messages.

A lighting device associated with a remote control device that is configured to be mounted overtop of a light switch (e.g., the remote control device **270** that is mounted over the toggle actuator **216** of the wall-mounted load control device **210**) may be assumed to be incapable of experiencing a local power removal event. For example, the light bulb **212** may be associated with the remote control device **270** for performing load control. The lighting bulb **212** may be assumed to be incapable of experiencing a local power removal event since the remote control device **270** retains the power state of the light bulb **212** in an on power state. Even when the remote control device **270** is actuated to change the light bulb **212** to the off power state, power may still be provided from the AC power source **202** to the light bulb **212** through the wall-mounted load control device **210**, which is maintained in the on position.

The light bulb **212** that is receiving power from the AC power source **202** through the wall-mounted load control device **210** having the remote control device **270** may experience a system power removal event. For example, the light bulb **212** may experience a blackout. As the remote control device **270** may prevent the light bulb **212** from experiencing a local power removal event, a power removal event experienced by the light bulb **212** coupled to the remote control device **270** may be determined to be a system power removal event. The light bulb **212** may determine that the power removal event experienced by the light bulb **212** is a system power removal event since the light bulb **212** is associated with the remote control device **270**. The hub device **280** may store the association of the light bulb **212** with the remote control device **270** and may determine that the power removal event experienced by the light bulb **212** is a system power removal event since the light bulb **212** is associated with the remote control device **270**. The light bulb **212** and/or the hub device **280** may control the state of the light bulb **212** based on the determination.

Before being associated with a remote control device (e.g., the battery-powered remote control device **250** and/or the remote control device **270**), the light bulb **212** may be configured to operate in an operation mode. The operation mode may be a switched mode (e.g., have stored in memory that the light bulb **212** should operate in the switched mode). When operating in the switched mode, the light bulb **212** may assume that each power removal event is a local power removal event and rely on the hub device **280** and/or other lighting devices to determine if the power removal event was a system power removal event. If the light bulb **212** is

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associated with the battery-powered remote control device **250** (e.g., that is not configured to be mounted over the toggle actuator **216** of the wall-mounted load control device **210**), the light bulb **212** may be configured to continue operating in the switched mode. If the light bulb **212** is associated with at least one remote control device that is configured to be mounted over the toggle actuator **216** of the wall-mounted load control device **210** (e.g., the remote control device **270**), the light bulb **212** may be configured to assume that the light bulb **212** is coupled downstream of a switch that is not able to be controlled to the off position. The unique identifier and/or a device type identifier of the battery-powered remote control device **250** and/or the remote control device **270** may be stored in memory to indicate the operation mode. The operation mode may be explicitly indicated in memory based on the association with the battery-powered remote control device **250** and/or the remote control device **270**. The light bulb **212** may store in memory that the light bulb **212** should operate in another operation mode, such as an unswitched mode, during which the light bulb **212** may assume that each power removal event is a system power removal event.

The light bulb **222** may experience a power removal event. The light bulb **222** may receive electrical power from the electrical receptacle **226** and/or the light bulb **222** may be controlled by another switch or from an AC power source other than the light bulb **212**. The light bulb **222** may be unassociated with the remote control device **270**. As the light bulb **222** is unassociated with remote control device **270**, the power removal event to the light bulb **222** may be determined to be a local power removal event (e.g., in response to an actuation of the actuator **225** of the table lamp **224**) or a system power removal event. To determine whether a power removal event is a local power removal event or a system power removal event, the power states of one or more lighting devices and/or the hub device **280** may be determined. For example, if the light bulb **222** is experiencing a power removal event and light bulb **212** is experiencing a power removal event, it may be determined that the power removal event is a system power removal event. The light bulb **222** may receive an indication from the light bulb **212** that the light bulb **212** is in an unswitched mode or associated with the remote control device **280** and use this indication to determine that the power removal event that was also experienced by the light bulb **212** was a system power removal event. If the light bulb **222** is experiencing a power removal event and lighting devices positioned within other rooms are experiencing a power removal event, it may be determined that the power removal event is a system power removal event. If the light bulb **222** is experiencing a power removal event and light bulb **212** is not experiencing a power removal event, it may be determined that the power removal event is a local power removal event.

Upon the lighting devices and/or the hub device **280** determining that the power removal event is a system power removal event, the power state of the lighting devices may be adjusted after the power removal event has ended. For example, after the power removal event has ended, the lighting devices and/or the hub device **280** may adjust the power state of the lighting devices to the prior power states of the lighting devices. For example, upon the lighting devices and/or the hub device **280** determining that the power removal event is a system power removal event, the lighting devices and/or the hub device **280** may obtain the prior power states of the lighting devices (e.g., via the memory of the lighting devices, the memory of the hub

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device **280**, the memory of the network device **290**, memory of an external server, etc.). The lighting devices and/or the hub device **280** may send and/or set the power states of the lighting devices to the obtained prior power states of the lighting devices.

FIG. 2D shows a load control system **200c** including additional control devices (e.g., input devices, load control devices, and/or controller devices). The load control system **200c** may include other load control devices, which may operate similar to other load control devices described herein. Examples of other load control devices may include a daylight control device, e.g., a motorized window treatment **230**, mounted in front of a window for controlling the amount of daylight entering the space in which the load control system **200c** is installed. The motorized window treatment **230** may include, for example, a cellular shade, a roller shade, a drapery, a Roman shade, a Venetian blind, a Persian blind, a pleated blind, a tensioned roller shade systems, or other suitable motorized window covering. The motorized window treatment **230** may include a motor drive unit **232** for adjusting the position of a covering material **234** of the motorized window treatment in order to control the amount of daylight entering the space. The motor drive unit **232** of the motorized window treatment **230** may have an RF receiver and an antenna mounted on or extending from a motor drive unit of the motorized window treatment. The motor drive unit **232** of the motorized window treatment **230** may be battery-powered or may receive power from an external direct-current (DC) power supply. Examples of battery-powered motorized window treatments are described in greater detail in commonly-assigned U.S. Pat. No. 8,950,461, issued Feb. 10, 2015, entitled *MOTORIZED WINDOW TREATMENT*, and U.S. Pat. No. 9,115,537, issued Aug. 25, 2015, entitled *BATTERY-POWERED ROLLER SHADE SYSTEM*, the entire disclosures of which are hereby incorporated by reference.

The load control system may include other input devices that may operate similar to the input devices described herein. For example, the load control system **200c** may include an occupancy sensor **260**. The occupancy sensor **260** may be configured to detect occupancy and vacancy conditions in the space in which the load control system **200c** is installed. The occupancy sensor **260** may transmit digital messages to the wall-mounted load control device **210** via the RF signals **206** in response to detecting the occupancy or vacancy conditions. The wall-mounted load control device **210** may be configured to turn on the light bulb **212** in response to receiving an occupied command. The wall-mounted load control device **210** may be configured to turn off the respective light bulb in response to receiving a vacant command. The occupancy sensor **260** may operate as a vacancy sensor to turn off (e.g., only turn off) the lighting devices in response to detecting a vacancy condition (e.g., to not turn on the light bulb **212** in response to detecting an occupancy condition). Examples of RF load control systems having occupancy and vacancy sensors are described in greater detail in commonly-assigned U.S. Pat. No. 8,009,042, issued Aug. 30, 2011, entitled *RADIO-FREQUENCY LIGHTING CONTROL SYSTEM WITH OCCUPANCY SENSING*; U.S. Pat. No. 8,199,010, issued Jun. 12, 2012, entitled *METHOD AND APPARATUS FOR CONFIGURING A WIRELESS SENSOR*; and U.S. Pat. No. 8,228,184, issued Jul. 24, 2012, entitled *BATTERY-POWERED OCCUPANCY SENSOR*, the entire disclosures of which are hereby incorporated by reference.

The load control system **200c** may include a daylight sensor **262**. The daylight sensor **262** may be configured to

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measure a total light intensity in the space in which the load control system is installed. The daylight sensor **262** may transmit digital messages including the measured light intensity to the wall-mounted load control device **210**. The daylight sensor **262** may transmit digital messages via the RF signals **206** for controlling the intensities of the light bulb **212** in response to the measured light intensity. Examples of RF load control systems having daylight sensors are described in greater detail in commonly-assigned U.S. Pat. No. 8,410,706, issued Apr. 2, 2013, entitled *METHOD OF CALIBRATING A DAYLIGHT SENSOR*; and U.S. Pat. No. 8,451,116, issued May 28, 2013, entitled *WIRELESS BATTERY-POWERED DAYLIGHT SENSOR*, the entire disclosures of which are hereby incorporated by reference.

The load control devices (e.g., the lightbulbs **212**, **222**, the motorized window treatment **220**, etc.) receiving the power removal event message and/or power state message may be two-way communication devices. For example, load control devices may receive messages (e.g., power removal event messages and/or power state messages), and/or may acknowledge receipt of the messages (e.g., power removal event messages and/or power state messages) to the transmitting device. The load control devices may identify the power removal event message and/or power state message as being from the hub device **280**, a lighting device, and/or the network device **290**. The load control devices may identify the hub device **280**, a lighting device, and/or a network device **290** by a device identifier (unique identifier).

The input devices (e.g., the occupancy sensor **260**, daylight sensor **262**, remote control device **250**, remote control device **270** etc.) receiving the power removal event message and/or power state message may be one-way or two-way communication devices. The input devices that are one-way communication devices may be unable to receive the power removal event message and/or the power state message. The one-way communication devices may transmit power state information and/or power removal event information. The power state information and/or the power removal event information may include the identifier of the transmitting device. To transmit the power state information and/or power removal event information, a button on the one-way communication device may be actuated or the device may identify a power startup at the device. To trigger the transmission of power state information and/or power removal event information at the daylight sensor **262**, a laser signal identifiable by the daylight sensor **262** may be transmitted. Though some control devices may be described as one-way or two-way communication devices, any control device may include a button for transmitting power state information and/or power removal event information.

The control devices (e.g., lighting devices, the battery-powered remote control device **250**, the occupancy sensor **260**, the daylight sensor **262**, the remote control device **270**, etc.), the hub device **280**, and/or the network device **290** may communicate with one another via one or more communication protocols. For example, the control devices (e.g., lighting devices, the battery-powered remote control device **250**, the occupancy sensor **260**, the daylight sensor **262**, the remote control device **270**, etc.) may be capable of communicating with the hub device **280** and/or the network device **290** via wireless signals (e.g., RF signals), such as WI-FI® signals; WIMAX® signals; BLUETOOTH® signals; near field communication (NFC) signals; proprietary communication signals, such as CLEAR CONNECT™; ZIGBEE® signals; Z-WAVE signals, and/or the like. Each device (e.g., each of the control devices, the hub device **280**, the network device **290**, etc.) may be capable of communicating on the

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same protocol and/or frequencies. Each device (e.g., each of the control devices, the hub device **280**, the network device **290**) may be capable of communicating on different protocols and/or frequencies. For example, the occupancy sensor **260** and/or the daylight sensor **262**, and the motorized window treatment **230** may communicate via one protocol or frequency (e.g., a proprietary protocol, such as CLEAR CONNECT™), the remote control device **270**, the battery-powered remote control device **250**, and/or the light bulbs **212**, **222** may communicate via one protocol or frequency (e.g., ZIGBEE®, BLUETOOTH, etc.), and/or the network device **290** may communicate via one protocol or frequency (e.g., WI-FI®, cellular, etc.). The occupancy sensor **260**, the daylight sensor **262**, and/or the motorized window treatment **230** may communicate using the same or different protocol or frequency as the remote control device **270**, the battery-powered remote control device **250**, and/or the light bulbs **212**, **222**. An intermediate device (e.g., the hub device **280**) may be used to allow communication between one or more devices communicating using one or more communication protocols.

The remote control device **270** may be coupled to and/or cover a switch (e.g., the toggle actuator **216** of the wall-mounted load control device **210**) as described above. The control devices (e.g., the remote control device **250**, the remote control device **270**, lightbulbs **212**, **222**, the occupancy sensor **260**, the daylight sensor **262**, the motorized window treatment **230**, etc.) may experience a power removal event. Control devices may not receive electrical power from a receptacle (e.g., electrical receptacle **226**), which is coupled to remote control device **270**. If other control devices are not associated with remote control device **270**, the power removal event to the other control devices may be determined to be a local power removal event and/or a system power removal event. To determine whether a power removal event is a local power removal event or a system power removal event, the power states of one or more of the control devices and/or the hub device **280** may be determined. For example, if the light bulb **212** is experiencing a power removal event and one or more of the other control devices is experiencing a power removal event, it may be determined that the power removal event is a system power removal event. If control devices positioned within one or more rooms are experiencing a power removal event, it may be determined that the power removal event is a system power removal event. If the light bulb **212** is not experiencing a power removal event and other control devices are experiencing a power removal event, it may be determined that the power removal event is a local power removal event. Lighting devices and/or the hub device **280** may determine the power states of one or more control devices.

FIG. 3 is a simplified flowchart depicting an example method **300** for retaining and/or adjusting power state information for lighting devices (e.g., the light bulb **212** and/or the light bulb **222**). The method **300** may depict an example method for retaining and/or adjusting power state information for lighting devices based on whether an operation mode (e.g., a switched mode) is set. As shown in FIG. 3, the method **300** may begin at **302**. At **304**, the lighting device and/or the hub device **280** may determine if the lighting device has encountered a power removal event. For example, the lighting device and/or the hub device **280** may determine if the lighting device has encountered a loss of power, such as blackout, a brownout, or another activity that has removed electrical power from the lighting device. The lighting device and/or the hub device **280** may determine

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information regarding the power removal event. For example, the lighting device and/or the hub device **280** may determine the time at which the power removal event occurred, the time at which power was last regained, the duration of the power removal event, and/or the duration since the power removal event ended.

If the lighting device determines that a power removal event has not occurred, the method may move to **306**. At **306**, the lighting device may determine whether power state information has been received. The power state information may include an on power state, an off power state, an intensity state, a color state, etc. For example, at **306**, the lighting device may determine whether the lighting device has received power state information of an intensity of sixty percent. The power state information may be received via one or more digital messages. For example, a lighting device may receive power state information from a control device, a network device, a hub device, etc.

If power state information is received at **306**, the lighting device may adjust its state, at **308**. The lighting device may adjust its state based on the received power state information. For example, a lighting device may set its state to an intensity of sixty percent if the device receives a digital message including a request for the lighting device to set its state to an intensity of sixty percent. The lighting device may, at **310**, store the received power state information. The lighting device may store the power state internally (e.g., in memory of the lighting device). The lighting device may store the power state externally (e.g., on an external device, such as another lighting device, a hub device, an external server, etc.). The method may return to **304** to determine if a power removal event has occurred.

If, at **304**, it is determined that a power removal event has occurred (e.g., the lighting device and/or the hub device **280** determines that a power removal event has occurred), the lighting device and/or the hub device **280** may determine, at **312**, whether an operation mode is set. For example, the lighting device and/or the hub device **280** may determine whether a switched mode is set and/or whether an unswitched mode is set. The switched mode may indicate that the lighting device is capable of being controlled by a switch (e.g., the toggle actuator **216** coupled to the wall-mounted load control device **210**). The unswitched mode may indicate that the lighting device is incapable of being controlled (e.g., turned off and/or turned on) by a switch, such as toggle actuator **216**. If, at **312**, the control is not set to the unswitched mode, the lighting device and/or the hub device **280** may set the power state of the lighting device to a predetermined state (e.g., an on power state at a predetermined intensity state) at **318**. For example, if the lighting device is not set to the unswitched mode, the lighting device and/or the hub device **280** may set the state of the lighting device to a full intensity level, a defined intensity state, and/or the prior power state. After setting the power state at **318**, the method may return to **304** to determine if a power removal event has occurred.

If, at **312**, the unswitched mode is set, the method **300** may move to **314**. At **314**, the lighting device and/or the hub device **280** may retrieve a power state of the lighting device. The power state of the lighting device may be a prior power state of the lighting device (e.g., a power state prior to the power removal event). For example, if the lighting device and/or the hub device **280** determines that a power removal event has occurred, at **304**, the lighting device and/or the hub device **280** may retrieve the power state that the lighting device and/or the hub device **280** previously and stored at **310**. At **316**, the lighting device may adjust its power state

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to the power state retrieved at **314**. As an example, at **316**, the lighting device may adjust its power state to an intensity of sixty percent if the prior power state was an intensity of sixty percent. After setting the power state at **316**, the method may return to **304** to determine if a power removal event has occurred.

FIG. **4** is a simplified flowchart depicting an example method **400** for retaining and/or adjusting state information (e.g., power state information) of one or more lighting devices (e.g., the light bulb **212** and/or the light bulb **222**). The method **400** may depict an example method for retaining and/or adjusting state information of one or more light devices, based on whether a power removal event is a local power removal event or a system power removal event. For example, method **400** may depict an example method for retaining and/or adjusting state information (e.g., power state information) of lighting devices using a hub device (such as the hub device **280**). As shown in FIG. **4**, the method **400** may begin at **402**. At **404**, the hub device may determine if the lighting device has encountered a power removal event. For example, the hub device may determine if the lighting device has encountered a blackout, a brownout, or another activity (such as a user physically turning off a switch to a lighting load) that has removed electrical power from the lighting device. The hub device may determine information regarding the power removal event. For example, the hub device may determine the time at which the power removal event occurred, the duration of the power removal event, and/or the duration since the power removal event ended.

If the hub device determines that a power removal event has not occurred, the method may move to **406**. At **406**, it may be determined whether power state information is received. The power state information may include information relating to the power state of the hub device **280**, and/or the power state information may include information relating to the power state of one or more lighting devices. The power state information may include an on power state, an off power state, an intensity state, a color state, etc. For example, at **406**, the lighting device may determine whether the hub device and/or the lighting device has received power state information of an intensity of sixty percent. The power state information may be received via one or more digital messages. For example, a lighting device may receive power state information from a control device, a network device, a hub device, etc.

If power state information is received at **406**, the received power state information may be stored at **408**. The lighting device may store the power state internally (e.g., in memory of the lighting device). The power state may be stored externally (e.g., on an external device, such as another lighting device, a hub device, an external server, etc.).

If power state information is received at **406**, the lighting device may adjust its power state at **410**. The lighting device may adjust its power state based on the received power state information. For example, a lighting device may set its state to an on power state if the lighting device receives a digital message including a request for the lighting device to set its state to an on power state. If the power state information is not received at **406**, or is received and the lighting device has performed adjustments, the method **400** may return to **404** to determine if a power removal event has occurred.

If, at **404**, it is determined that a power removal event has occurred (e.g., the hub device **280** determines that a power removal event has occurred), the hub device **280** may determine, at **412**, whether the power removal event is a system power removal event, at **412**. The hub device **280**

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may determine that a power removal event is a system power removal event (e.g., a brownout and/or a blackout) by querying lighting devices positioned at different locations and/or by querying lighting devices that are within groups positioned at different locations. For example, the hub device **280** may query lighting devices connected to different circuits (e.g., AC circuits) to determine if the power removal event is a system power removal event (e.g., a power removal event affecting more than one AC circuit, such as a brownout or blackout) and/or whether the power removal event is a local power removal event (e.g., a power removal event affecting a single AC circuit, such as when a user physically turns off a switch to a lighting device, for example).

If, at **412**, the power removal event is determined to not be a system power removal event, the lighting device may set the state of the control device to a predetermined state (e.g., an on power state at a predetermined intensity state), at **418**. For example, if the power removal event is determined to not be a system power removal event, the lighting device may set the state of the lighting device to a full intensity level, a defined intensity state, and/or the prior power state. After setting the power state at **418**, the method may return to **404** to determine if a power removal event has occurred.

If, at **412**, the power removal event is determined to be a system power removal event, move to **414**. At **414**, the lighting device may retrieve a power state of the lighting device. The power state of the lighting device may be a prior power state of the lighting device (e.g., a power state prior to the power removal event). For example, if the lighting device and/or the hub device **280** determines that a system power removal event has occurred, at **410**, the lighting device and/or the hub device **280** may retrieve the power state that the lighting device previously stored at **410**. At **416**, the lighting device may adjust its power state to the power state retrieved at **414**. As an example, at **416**, the lighting device may adjust its power state to an intensity of sixty percent if the prior power state was an intensity of sixty percent. After setting the power state at **416**, the method may return to **404** to determine if a power removal event has occurred.

FIG. **5** is a simplified flowchart depicting an example method **500** for associating a lighting device (e.g., the light bulb **212** and/or the light bulb **222**) with a retrofit remote control device (e.g., the retrofit remote control device **270**). For example, the method **500** may be executing by the lighting device for determining whether state information (e.g., power state information) of one or more control devices is to be adjusted and/or retained, based on the use of the retrofit remote control device **270**. The retrofit remote control device may be configured to be coupled to a toggle actuator of a light switch (e.g., the toggle actuator **216** of the wall-mounted load control device **210**) that is coupled in series electrical connection with the lighting device. The retrofit remote control device may be configured to retain the light switch in an on position. The retrofit remote control device may communicate with the lighting device, for example, via one or more of the devices via wireless signals, e.g., radio-frequency (RF) signals **206** (e.g., ZIGBEE®, NFC; BLUETOOTH®, WI-FI®, or a proprietary communication channel, such as CLEAR CONNECT™, etc.).

As shown in FIG. **5**, the method **500** may begin at **502**. At **504**, an association message may be received. For example, at **504**, the lighting device may receive an association message. The association message may be sent by one or more devices. For example, the association message may be

sent by control devices (e.g., lighting devices, remote control devices, etc.) and/or hub devices.

The association message may be transmitted from a device in an association mode. The association message may indicate a unique identifier of the device and/or a device type from which the association message is transmitted. For example, an association message from a lighting device may indicate that the message is from a lighting device, an association message from a wall-mounted dimmer may indicate that the message is from a wall-mounted dimmer, and/or an association message from a retrofit remote control device may indicate that the message is from a retrofit remote control device. The device type may be indicated by the unique identifier, or a separate identifier. The association message may indicate the location of the device in the load control system from which the association message is received. The association message may identify the devices that are coupled to and/or associated with the device from which the association message is sent. For example, a device may send association information, including the identification of devices and/or device types, stored on the device during association and from which the device may receive digital messages for performing control.

At **506**, it may be determined if the association message was received from a retrofit remote control device. The association message may indicate the other control devices with which the retrofit device is associated. For example, if the lighting device receives an association message from a retrofit remote control device during an association mode, the lighting device may be associated with the retrofit device. The lighting device may also identify the retrofit remote control device as being previously associated with another lighting device from association information indicating previous associations. Based on the association information, the lighting device may determine the location of the retrofit remote control device and/or the other control devices (e.g., lighting devices) to which the retrofit remote control device is associated.

If, at **506**, it is determined that a retrofit device is not being used, the method **500** may move to **510**. If, at **506**, it is determined that a retrofit device is being used, an operation mode may be set at **508** (e.g., an unswitched mode may be set). The unswitched mode may indicate that the lighting device is incapable of being controlled (e.g., turned off and/or turned on) by a switch (e.g., the wall-mounted load control device **210** shown in FIG. 2). For example, the unswitched mode may indicate that the lighting device is not downstream of a light switch that is capable of being switched off at the time. The unswitched mode may indicate that the lighting device is downstream of a light switch to which a retrofit remote control device is mounted, where the retrofit remote control device may retain the light switch in the on position.

At **510**, association information received in the association message at **504** may be stored. For example, at **510**, a unique identifier (e.g., serial number) of the retrofit remote control device and/or the associated control devices (e.g., a lighting device, such as light bulb **212**) may be stored. The association information may include the unique identifier (e.g., serial number) of other devices, the device types, locations, etc. of devices. The association information may be stored by the hub device **280**, a lighting device, and/or the network device **290**. The association information may be stored so that the hub device **280**, the lighting devices, and/or the network device **290** may determine the control devices that are associated with a retrofit remote control device. The association information may be stored so that

the hub device **280**, the lighting devices, and/or the network device **290** may identify the control devices that are unassociated with a retrofit remote control device.

If a control device (e.g., lighting device) is associated with a retrofit remote control device, the control device may be determined to have experienced a system power removal event if the control device experiences a power removal event. For example, if the light bulb **212** is associated with the retrofit remote control device **270** and the light bulb **212** experiences a power removal event, it may be determined that the light bulb **212** has experienced a system power removal event. The light bulb **212** may be determined to have experienced a system power removal event because the retrofit remote control device **270** is configured to maintain the toggle actuator **216** of the wall-mounted remote control device **210** in the on position and the power state of the light bulb **212** in the on power state. For example, if the retrofit remote control device **270** is configured to prevent a user from actuating the toggle actuator **216** in the on position and the wall-mounted remote control device from turning off the light bulb **212**, the light bulb **212** may be turned to an off power state during a system power removal event (e.g., brownout, blackout, etc.). The light bulb **212** may be prevented from being turned off based on a local power removal event (e.g., turning the toggle actuator **216** to the off position). If the light bulb **212**, other lighting devices, and/or the hub device **280** determines that the light bulb **212** has experienced a power removal event, the light bulb **212**, the other lighting devices, and/or the hub device **280** may determine that the power removal event of the light bulb **212** is a system power removal event. The method may end at **512**.

FIG. 6 is a block diagram illustrating an example control device **600**, e.g., a load control device, as described herein. The control device **600** may be a lighting device (e.g., the light bulb **212**, such as shown in FIG. 2). The control device **600** may be a dimmer device, an electronic switch, an electronic ballast for lamps, an LED driver for LED light sources, an AC plug-in control device, a temperature control device (e.g., a thermostat), a motor drive unit for a motorized window treatment, or other control device. The control device **600** may comprise a load regulation circuit **608** that may control an electrical load **616**. The electrical load **616** may include any type of electrical load. For example, the electrical load may be a lighting load and/or a lighting device. The load regulation circuit **608** may receive power via a hot connection **612** and a neutral connection **614** (e.g., for connection to a power source, such as an AC power source) and may provide an amount of power to the electrical load **616**. When the control device **600** is a light bulb (e.g., such as the light bulb **212** and/or the light bulb **222**), the electrical load **616** may be integral to the control device **600**. When the control device **600** is a light bulb and the electrical load is, for example, an LED light source, the load regulation circuit **608** may comprise an LED drive circuit. When the control device **600** is a dimmer device (e.g., such as dimmer device **218**, shown in FIG. 2B), the electrical load **616** may be external to the control device (e.g., as shown in FIG. 6) and the control device may be coupled to the electrical load via a single electrical connection. The electrical load **616** may be coupled to the neutral side of the AC power source, and the control device **600** may not include the neutral connection **614** to the neutral side of the AC power source. When the control device **600** is a dimmer device, the load regulation circuit **608** may comprise a dimming circuit for controlling the power delivered to the electrical load using a phase-control technique. The control

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circuit 604 may determine the status of the electrical load 616 in response to status feedback received from the load regulation circuit 608.

The control device 600 may comprise a control circuit 604 that may control the load regulation circuit 608 for controlling the electrical load 616. The control circuit 604 may include one or more general purpose processors, special purpose processors, conventional processors, digital signal processors (DSPs), microprocessors, integrated circuits, a programmable logic device (PLD), application specific integrated circuits (ASICs), or the like. The control circuit 604 may perform signal coding, data processing, power control, input/output processing, or any other functionality that enables the control device 600 to perform as described herein. The control device 600 may include a communications circuit 602. The communications circuit 602 may include an RF transceiver or other communications module capable of performing wired and/or wireless communications via a communications link (e.g., a wireless or wired communication link). The communications circuit 602 may be in communication with the control circuit 604.

The control circuit 604 may store information in and/or retrieve information from the memory 606. For example, the memory 606 may maintain a registry of associated control devices and/or control configuration instructions. The memory 606 may include a non-removable memory and/or a removable memory.

The control circuit 604 may be in communication with actuators 618 (e.g., one or more buttons) that may be actuated by a user to communicate user selections to the control circuit 604. For example, the actuators 618 may be actuated to put the control circuit 604 in an association mode and/or communicate association messages from the control device 600.

FIG. 7 is a block diagram illustrating an example input device 700 as described herein. The input device 700 may be a remote control device (such as remote control device 250 and/or remote control device 270), an occupancy sensor, a daylight sensor, a window sensor, a temperature sensor, and/or the like. The input device 700 may include a control circuit 702 for controlling the functionality of the input device 700. The control circuit 702 may include one or more general purpose processors, special purpose processors, conventional processors, digital signal processors (DSPs), microprocessors, integrated circuits, a programmable logic device (PLD), application specific integrated circuits (ASICs), or the like. The control circuit 702 may perform signal coding, data processing, power control, input/output processing, or any other functionality that enables the input device 700 to perform as described herein.

The control circuit 702 may store information in and/or retrieve information from the memory 704. The memory 704 may include a non-removable memory and/or a removable memory, as described herein.

The input device 700 may include a communications circuit 708 for transmitting and/or receiving information. The communications circuit 708 may transmit and/or receive information via wired and/or wireless communications. The communications circuit 708 may include a transmitter, an RF transceiver, or other circuit capable of performing wired and/or wireless communications. The communications circuit 708 may be in communication with control circuit 702 for transmitting and/or receiving information.

The control circuit 702 may also be in communication with an input circuit 706. The input circuit 706 may include an actuator (e.g., one or more buttons) or a sensor circuit

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(e.g., an occupancy sensor circuit, a daylight sensor circuit, or a temperature sensor circuit) for receiving input that may be sent to a device for controlling an electrical load. For example, the input device 700 may receive input from the input circuit 706 to put the control circuit 702 in an association mode and/or communicate association messages from the input device 700. The control circuit 702 may receive information from the input circuit 706 (e.g., an indication that a button has been actuated or sensed information). Each of the modules within the network device 700 may be powered by a power source 710.

FIG. 8 is a block diagram illustrating an example network device 800 as described herein. The network device 800 may include the network device 290, shown in FIGS. 2A-2D, for example. The network device 800 may include a control circuit 802 for controlling the functionality of the network device 800. The control circuit 802 may include one or more general purpose processors, special purpose processors, conventional processors, digital signal processors (DSPs), microprocessors, integrated circuits, a programmable logic device (PLD), application specific integrated circuits (ASICs), or the like. The control circuit 802 may perform signal coding, data processing, power control, input/output processing, or any other functionality that enables the network device 800 to perform as described herein. The control circuit 802 may store information in and/or retrieve information from the memory 804. The memory 804 may include a non-removable memory and/or a removable memory. The non-removable memory may include random-access memory (RAM), read-only memory (ROM), a hard disk, or any other type of non-removable memory storage. The removable memory may include a subscriber identity module (SIM) card, a memory stick, a memory card, or any other type of removable memory.

The network device 800 may include a communications circuit 808 for transmitting and/or receiving information. The communications circuit 808 may perform wireless and/or wired communications. The communications circuit 808 may include an RF transceiver or other circuit capable of performing wireless communications via an antenna. Communications circuit 808 may be in communication with control circuit 802 for transmitting and/or receiving information.

The control circuit 802 may also be in communication with a display 806 for providing information to a user. The processor 802 and/or the display 806 may generate GUIs for being displayed on the network device 800. The display 806 and the control circuit 802 may be in two-way communication, as the display 806 may include a touch screen module capable of receiving information from a user and providing such information to the control circuit 802. The network device may also include an actuator 812 (e.g., one or more buttons) that may be actuated by a user to communicate user selections to the control circuit 802.

Each of the modules within the network device 800 may be powered by a power source 810. The power source 810 may include an AC power supply or DC power supply, for example. The power source 810 may generate a supply voltage V_{CC} for powering the modules within the network device 800.

FIG. 9 is a block diagram illustrating an example hub device 900 as described herein. The hub device 900 may include a control circuit 902 for controlling the functionality of the hub device 900. The control circuit 902 may include one or more general purpose processors, special purpose processors, conventional processors, digital signal processors (DSPs), microprocessors, integrated circuits, a pro-

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grammable logic device (PLD), application specific integrated circuits (ASICs), or the like. The control circuit **902** may perform signal coding, data processing, power control, input/output processing, or any other functionality that enables the hub device **900** to perform as described herein. The control circuit **902** may store information in and/or retrieve information from the memory **904**. The memory **904** may include a non-removable memory and/or a removable memory. The non-removable memory may include random-access memory (RAM), read-only memory (ROM), a hard disk, or any other type of non-removable memory storage. The removable memory may include a subscriber identity module (SIM) card, a memory stick, a memory card, or any other type of removable memory.

The hub device **900** may include a communications circuit **906** for transmitting and/or receiving information. The communications circuit **906** may perform wireless and/or wired communications. The hub device **900** may also, or alternatively, include a communications circuit **908** for transmitting and/or receiving information. The communications circuit **906** may perform wireless and/or wired communications. Communications circuits **906** and **908** may be in communication with control circuit **902**. The communications circuits **906** and **908** may include RF transceivers or other communications modules capable of performing wireless communications via an antenna. The communications circuit **906** and communications circuit **908** may be capable of performing communications via the same communication channels or different communication channels. For example, the communications circuit **906** may be capable of communicating (e.g., with a network device, over a network, etc.) via a wireless communication channel (e.g., BLUETOOTH®, near field communication (NFC), WIFI®, WI-MAX®, cellular, etc.) and the communications circuit **908** may be capable of communicating (e.g., with control devices and/or other devices in the load control system) via another wireless communication channel (e.g., WI-FI® or a proprietary communication channel, such as CLEAR CONNECT™).

The control circuit **902** may be in communication with an LED indicator **912** for providing indications to a user. The control circuit **902** may be in communication with an actuator **914** (e.g., one or more buttons) that may be actuated by a user to communicate user selections to the control circuit **902**. For example, the actuator **914** may be actuated to put the control circuit **902** in an association mode and/or communicate association messages from the hub device **900**.

Each of the modules within the hub device **900** may be powered by a power source **910**. The power source **910** may include an AC power supply or DC power supply, for example. The power source **910** may generate a supply voltage V_{CC} for powering the modules within the hub device **900**.

Although features and elements are described herein in particular combinations, each feature or element can be used alone or in any combination with the other features and elements. The methods described herein may be implemented in a computer program, software, or firmware incorporated in a computer-readable medium for execution by a computer or processor. Examples of computer-readable media include electronic signals (transmitted over wired or wireless connections) and computer-readable storage media. Examples of computer-readable storage media include, but are not limited to, a read only memory (ROM), a random access memory (RAM), removable disks, and optical media such as CD-ROM disks, and digital versatile disks (DVDs).

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What is claimed is:

1. A device comprising:

a communication circuit configured to transmit messages to and receive messages from at least one control device;

a memory; and

a control circuit configured to:

detect a loss of power from a first power source;

in response to the loss of power from the first power source, receive power from a second power source; store, in the memory, an indication of the loss of the power from the first power source;

determine, based on the loss of power from the first power source and at least one first message communicated between the device and the at least one control device, that the loss of power from the first power source is a result of a system power removal event; and

in response to the determination that the loss of power is the result of the system power removal event, transmit, via the communication circuit, at least one second message to the at least one control device that is configured to cause the at least one control device to return to a prior power state before the system power removal event.

2. The device of claim 1, wherein the control circuit is configured to receive the at least one first message from the at least one control device, wherein the at least one first message indicates that the at least one control device detected a loss of power.

3. The device of claim 2, wherein the control circuit is configured to receive the at least one first message from the at least one control device without the at least one first message being requested.

4. The device of claim 2, wherein the control circuit is configured to:

transmit, via the communication circuit based on the return of the power from the first power source, at least one third message to the at least one control device, wherein the at least one third message is configured as a request for whether the at least one control device detected the loss of power; and

receive, via the communication circuit, the at least one first message in response to the at least one third message.

5. The device of claim 4, wherein the at least one control device comprises a plurality of control devices, wherein the at least one first message comprises a plurality of first messages that are each configured to be received from a respective control device of the plurality of control devices.

6. The device of claim 5, wherein the control circuit is configured to determine, based on a threshold number of the plurality of control devices indicating the loss of power in the plurality of first messages, that the loss of power from the first power source is the result of the system power removal event.

7. The device of claim 5, wherein the control circuit is configured to determine the loss of power from the first power source to be the result of the system power removal event based on each of the plurality of control devices indicating the detection of the loss of power in the plurality of first messages.

8. The device of claim 1, wherein the second power source is a battery, and wherein the device comprises the battery.

9. The device of claim 1, wherein the indication comprises a time at which the loss of power was detected, wherein the

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at least one first message comprises a time at which a loss of power was detected at the at least one control device, and wherein the control circuit is configured to determine that the loss of power is the result of the system power removal event based on a comparison of the time at which the loss of power was detected at the device to the time at which the loss of power was detected at the at least one control device.

10. The device of claim 1, wherein the loss of power is a first loss of power, the control circuit configured to:

receive at least one third message from the at least one control device indicating that the at least one control device experienced a second loss of power;

determine, based on the device failing to detect a second loss of power from the first power source at a time at which the at least one control device experienced the second loss of power, that the second loss of power is a result of a local power removal event; and

in response to the determination that the second loss of power is the result of the local power removal event, transmit, via the communication circuit, at least one fourth message to the at least one control device that is configured to indicate to the at least one control device that the second loss of power is the result of the local power removal event.

11. The device of claim 1, wherein the control circuit is configured to:

receive, via the communication circuit, a third message from an input device that is configured to control a state of the at least one control device; and

store the state of the at least one control device in the memory, wherein the state is used as the prior power state transmitted in the at least one second message in response to the determination that the loss of power is the result of the system power removal event.

12. The device of claim 1, wherein the control circuit is configured to:

detect a return of the power from the first power source; and

wherein the control circuit is configured to transmit, via the communication circuit, the at least one second message after the detection of the return from the first power source.

13. The device of claim 1, wherein the control circuit is configured to:

detect a return of the power from the first power source; and

wherein the control circuit is configured to determine, after the return of the power from the first power source, that the loss of power from the first power source is the result of the system power removal event.

14. A method comprising:

detecting a loss of power from a first power source at a device;

in response to the loss of power from the first power source at the device, receiving power from a second power source at the device;

storing, in memory, an indication of the loss of the power from the first power source at the device;

determining, based on the loss of power from the first power source at the device and at least one first message communicated between the device and at least one control device, that the loss of power from the first power source is a result of a system power removal event; and

in response to the determination that the loss of power is the result of the system power removal event, transmitting at least one second message to the at least one

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control device that is configured to cause the at least one control device to return to a prior power state before the system power removal event.

15. The method of claim 14, wherein the at least one first message is received from the at least one control device, wherein the at least one first message indicates that the at least one control device detected a loss of power.

16. The method of claim 15, further comprising:

transmitting, based on the detection of the return of the power from the first power source, at least one third message to the at least one control device, wherein the at least one third message is configured as a request for whether the at least one control device detected the loss of power; and

receiving the at least one first message in response to the at least one third message.

17. The method of claim 16, wherein the at least one control device comprises a plurality of control devices, wherein the at least one first message comprises a plurality of first messages that are each configured to be received from a respective control device of the plurality of control devices.

18. The method of claim 17, wherein the loss of power from the first power source is determined to be the result of the system power removal event based on each of the plurality of control devices indicating the detection of the loss of power in the plurality of first messages.

19. The method of claim 14, wherein the indication comprises a time at which the loss of power was detected, wherein the at least one first message comprises a time at which a loss of power was detected at the at least one control device, and wherein the loss of power is determined to be the result of the system power removal event based on a comparison of the time at which the loss of power was detected at the device to the time at which the loss of power was detected at the at least one control device.

20. The method of claim 14, wherein the loss of power is a first loss of power, the method further comprising:

receiving at least one third message from the at least one control device indicating that the at least one control device experienced a second loss of power;

determining, based on the device failing to detect a second loss of power from the first power source at a time at which the at least one control device experienced the second loss of power, that the second loss of power is a result of a local power removal event; and

in response to the determination that the second loss of power is the result of the local power removal event, transmitting at least one fourth message to the at least one control device that is configured to indicate to the at least one control device that the second loss of power is the result of the local power removal event.

21. The method of claim 14, further comprising:

receiving a third message from an input device that is configured to control a state of the at least one control device; and

storing the state of the at least one control device in the memory, wherein the state is used as the prior power state transmitted in the at least one second message in response to the determination that the loss of power is the result of the system power removal event.

22. The method of claim 14, further comprising:

detecting a return of the power from the first power source; and

wherein the loss of power from the first power source is determined to be the result of the system power

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removal event after the return of the power from the first power source is detected.

23. At least one computer-readable storage medium having instructions stored thereon that are configured to, when executed by at least one control circuit, cause the at least one control circuit to:

detect a loss of power from a first power source at a device;

in response to the loss of power from the first power source at the device, receive power from a second power source at the device;

store, in memory, an indication of the loss of the power from the first power source at the device;

determine, based on the loss of power from the first power source at the device and at least one first message communicated between the device and at least one control device, that the loss of power from the first power source is a result of a system power removal event; and

in response to the determination that the loss of power is the result of the system power removal event, transmit at least one second message to the at least one control device that is configured to cause the at least one control device to return to a prior power state before the system power removal event.

24. The at least one computer-readable storage medium of claim 23, wherein the instructions are configured to cause the at least one control circuit to receive the at least one first message from the at least one control device, wherein the at least one first message indicates that the at least one control device detected a loss of power.

25. The at least one computer-readable storage medium of claim 24, the instructions being further configured to cause the at least one control circuit to:

transmit, based on the detection of the return of the power from the first power source, at least one third message to the at least one control device, wherein the at least one third message is configured as a request for whether the at least one control device detected the loss of power; and

receive the at least one first message in response to the at least one third message.

26. The at least one computer-readable storage medium of claim 25, wherein the at least one control device comprises a plurality of control devices, wherein the at least one first message comprises a plurality of first messages that are each configured to be received from a respective control device of the plurality of control devices.

27. The at least one computer-readable storage medium of claim 26, wherein the instructions are configured to cause the at least one control circuit to determine the loss of power from the first power source to be the result of the system

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power removal event based on each of the plurality of control devices indicating the detection of the loss of power in the plurality of first messages.

28. The at least one computer-readable storage medium of claim 23, wherein the indication comprises a time at which the loss of power was detected, wherein the at least one first message comprises a time at which a loss of power was detected at the at least one control device, and wherein the loss of power is determined to be the result of the system power removal event based on a comparison of the time at which the loss of power was detected at the device to the time at which the loss of power was detected at the at least one control device.

29. The at least one computer-readable storage medium of claim 23, wherein the loss of power is a first loss of power, the instructions being further configured to cause the at least one control circuit to:

receive at least one third message from the at least one control device indicating that the at least one control device experienced a second loss of power;

determine, based on the device failing to detect a second loss of power from the first power source at a time at which the at least one control device experienced the second loss of power, that the second loss of power is a result of a local power removal event; and

in response to the determination that the second loss of power is the result of the local power removal event, transmit at least one fourth message to the at least one control device that is configured to indicate to the at least one control device that the second loss of power is the result of the local power removal event.

30. The at least one computer-readable storage medium of claim 23, the instructions being further configured to cause the at least one control circuit to:

receive a third message from an input device that is configured to control a state the at least one control device; and

store the state of the at least one control device in the memory, wherein the state is used as the prior power state transmitted in the at least one second message in response to the determination that the loss of power is the result of the system power removal event.

31. The at least one computer-readable storage medium of claim 23, the instructions being further configured to cause the at least one control circuit to:

detect a return of the power from the first power source; and

wherein the loss of power from the first power source is determined to be the result of the system power removal event after the return of the power from the first power source is detected.

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